



**SKYMAX AVIATION LIMITED,
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FLIGHT RADIOTELEPHONY OPERATOR'S LICENSE



FIFE YANKEE
SIERRA ALPHA
LIMA

CHAPTER 1

FLIGHT RADIOTELEPHONY OPERATOR LICENCE

1.1 General eligibility requirements

- (1) Except for a holder of a pilot licence, a person required to use radiotelephony apparatus aboard an aircraft must hold a flight radiotelephony operator licence.
- (2) An applicant for a radiotelephony operator licence shall:
 - (a) Be at least 17 years of age;
 - (b) Demonstrate the ability to read, speak write and understand the English language in accordance with the language proficiency requirements;
 - (c) Comply with the knowledge and skill requirements, for flight radiotelephony operator; and
 - (d) Demonstrate a level of knowledge appropriate to the privileges granted to the holder of the licence.

1.2 Skill and knowledge requirements

- (1) An applicant for the licence shall pass a practical and knowledge test covering the following areas:
 - (a) International Civil Aviation Organization (ICAO) spelling alphabet;
 - (b) Departure and position reporting;
 - (c) Obtaining meteorological information;
 - (d) Transmission and procedures of distress and urgency signals;
 - (e) Communication technique procedures;
 - (f) The necessity for brevity in radiotelephony communication and priorities;
 - (g) Pre-flight briefing;
 - (h) Classification of directional finding bearings;
 - (i) Radiotelephony facilities and frequencies available in the flight information region;
 - (j) Elementary knowledge of the relationship between wavelength and frequency;
 - (k) Radiotelephony procedures and phraseology; and
 - (l) The ability to use the radio equipment of the type installed in the aircraft and including the ability to carry out emergency procedures.
- (2) The results of the knowledge test of the applicant for the licence shall be valid for six months after the applicant passes the test

1.3 **Authority to Issue Licence**

Director General, Kenya Civil Aviation Authority is the authority responsible for the issue of flight radiotelephony operator's license.

Licenses are issued to persons who qualify in accordance with requirements laid down in CAP 90 "The Flight Radiotelephony Operator's license," and as stated in 1.2 above.

1.4 **Licensing of Aircraft Radio Station**

The authority for approving the design and installation of radio equipment in an aircraft is the Director General of civil aviation. This approval must be in writing, while the Commissioner General of the Communications Commission of Kenya (CCK.) is the authority for licensing the radio station.

1.5 **Application for RTF License.**

The following are the requirements to be met by a RTF applicant

- (i) Complete application forms obtainable from the licensing section of the KCAA No medical examination is required.
- (ii) Photos:- Complete application forms forwarded to licensing section with two un-mounted photographs of the head (1¼" x ¾").

1.6 **Fees**

Since the fees are liable to variations, reference for chargeable fees should be made to the latest AIC.

NOTES:

- (i) License may be renewed by just paying appropriate fee.
- (ii) No renewal fee is charged where the holder also holds a current pilot's licence.
- (iii) Where a practical exam is required a reduced fee may be charged.

1.7 **Privilege of Licence**

The holder of the licence shall have the privilege to use the radiotelephone on board an aircraft. He/she must sign his/her name on the licence in ink in the space provided.

1.8 Radiotelephony Licensing Examination

The examination is divided into two parts; a practical and a written paper of 30 minutes duration. Both parts must be passed within a period of six months and the standard pass mark in each part is 90%.

1.9 Exemptions from Examinations

Exemption from RTF licensing examination may be granted to:

- (a) Applicants who provide evidence of possession or equivalent qualification.
- (b) Holders of a Kenya ATC Licenses.

1.10 Conversion of Foreign Licence

To convert any foreign pilot's licence which does not include RTF Operator's Licence, the applicant must sit for and pass a RTF examination before such licence can be converted to Kenya Licence.

1.11 English Proficiency Test

It is an ICAO requirement that every pilot be tested and issued an English Proficiency Certificate. The test comprises of the following:

- (a) About 5 minute interview on why you are working or have interest in aviation.
- (b) About 10 minute conversation about aviation in general.
- (c) About 10 minutes listening to a taped ATC conversation (various nationality accents – American, Arabic, Spanish, etc.) and selecting correct responses from choices (A, B, C, D) given.

1.12 Maximum period of validity

The maximum validity period for a R/T licence is 24 months.

1.13 Renewal Requirements

A holder of the licence may apply for renewal of the licence if he/she has exercised its privileges in the six months preceding the date of application.

CHAPTER 2

AIR TRAFFIC SERVICES

2.1 Definitions

Aerodrome: A defined area on land or water (including any buildings, installations and equipment) intended to be used either wholly or in part for the arrival, departure and surface movements of aircraft.

Aerodrome Control Service: Air traffic control service for aerodrome traffic.

Aerodrome Taxi Circuit: The specified path of aircraft on the manoeuvring area during specific wind conditions.

Aerodrome Traffic: All traffic on the manoeuvring area of an aerodrome and all aircraft flying in the vicinity of an aerodrome.

Aerodrome Traffic Circuit: The specified path to be flown by aircraft operating in the vicinity of an aerodrome.

Air Traffic Advisory Service: A service provided within advisory airspace to ensure separation, in so far as possible, between aircraft operating on IFR, flight space.

Air Traffic Control Service: A service provided for the purpose of:

- (1) Preventing collisions: (i) between aircraft in the air; and
(ii) on the maneuvering area, between aircraft and obstructions.
- (2) Expediting and maintaining orderly flow of air traffic.

Air Traffic Services: This is a generic term meaning variously flight information service, alerting service, air traffic advisory service and air traffic control service.

Aerodrome Traffic Zone: An airspace of defined dimensions established around an aerodrome for the protection of aerodrome traffic.

Alerting Service: A service provided to notify appropriate organization regarding aircraft in need of SAR and to assist such organization as required.

Altitude: The vertical distance of a level, a point or an object considered as a point, measured from mean sea level.

Blind Transmission: A transmission from one station to another station in circumstances where two-way communication cannot be established, but where it is believed that the called station is able to receive the transmission.

Broadcast: A transmission of information relating to air navigation that is not addressed to a specific station or stations.

Flight Information Service: A service provided for the purpose of giving advice and information useful for the safe and efficient conduct of flights.

Flight Level: A surface of constant atmospheric pressure which is related to a specific datum, 1013.25 hPa, and is separated from other such surfaces by specific pressure intervals.

Height: The vertical distance of a level, a point or an object considered as a point, measured from a specified datum.

Level: A generic term relating to the vertical position of an aircraft in flight and meaning variously height, altitude and flight level.

Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft excluding aprons.

Radar: A radio detection device which provides information on range, azimuth and elevation of objects.

Secondary Surveillance Radar (SSR): A radar surveillance system which uses transmitters/receivers (interrogators) and transponders.

Transition Altitude: An altitude at or below which the vertical position of an aircraft is controlled by reference to altitudes.

Transition Layer: The airspace between the transition altitude and the transition layer.

Transition Level: The lowest flight level available for use above the transition altitude.

2.2 Units of Measurement

The International System of Units (SI Units) developed and maintained by the General Conference of Weights and Measures (CGPM) shall be used as the standard system of units of measurement for all aspects of international civil aviation air and ground operations.

Some non-SI Units are considered for permanent use with the SI Units. These shall be used either in lieu of, or in addition to, SI Units as primary units. Such units are: feet, nautical miles, knots, feet per minute.

The table of units shown below is used by the Aeronautical stations in the Nairobi FIR.

<i>For measurement of:</i>	<i>Units used</i>
<i>Distance used in navigation, position reporting, etc. – generally in excess of 2 nautical miles</i>	<i>*Nautical Miles and tenths</i>
<i>Relatively short distances such as those relating to aerodrome (e.g. runway lengths)</i>	<i>Metres</i>
<i>+ Altitudes, Elevation and Heights</i>	<i>Feet/Metres</i>
<i>Horizontal speed including wind speed</i>	<i>Knots/Km per hour</i>
<i>Vertical speed</i>	<i>Feet per minute</i>
<i>Wind direction for landing and taking off</i>	<i>Degrees Magnetic</i>
<i>Wind direction except for landing and taking off</i>	<i>Degrees True</i>
<i>Visibility including runway visual range</i>	<i>Kilometres or Metres</i>
<i>Altimeter setting</i>	<i>Hectopascal</i>
<i>Temperature</i>	<i>Degrees Celsius (Centigrade)</i>
<i>Weight</i>	<i>Metric Tonnes or Kilograms</i>
<i>Time</i>	<i>Hours and minutes, the day of 24 hours beginning at midnight UTC</i>
<i>Notes:</i> <i>1. * One international nautical mile = 1,852 Metres</i> <i>2. + Cruising levels are given in flight level numbers and not in altitudes</i>	

2.3 Time System

Coordinated Universal Time (UTC) is used by the air navigation services. This is 3 hours behind the East African Local Time

2.4 Units of ATC

The units which provide Air Traffic Control Services include:

- (i) Area Control Service (ACC),
- (ii) Radar Control Service (RADAR),
- (iii) Approach Control Service (APPROACH), and
- (iv) Aerodrome Control Service (TOWER)

2.5 Use of SSR for ATC Services

All aircraft in the Nairobi FIR are required to carry serviceable Secondary Radar Transponder with either Mode A, Mode C or Mode S capable of replying to interrogations from Secondary Radar installations on the ground and set in accordance with instructions from the appropriate Radar Units.

Validity checks will be requested by Radar Controllers to verify the accuracy of level information relayed. In cases where the information shows inaccuracy of more than 300ft, the pilot will be advised to cease transmitting on Mode C.

2.6 Special SSR Codes

In the current Mode A and Mode C Transponders, there are a total possible combination for 4096 codes.

SSR Transponder Codes may be used to relay or convey specific special messages. Some of important codes to remember are:

<u>Code</u>	<u>To Be Selected When:</u>
2000	Flying under IFR until assigned specific code by ATS Unit.
2077	Flying under VFR until assigned specific code by ATS Unit.
7500	Aircraft subject to unlawful interference (Hijacked)
7600	Aircraft experiencing two-way radio communication failure.
7700	Aircraft flying in a state of emergency.

(See standard phraseology for SSR communications at APPENDIX 3)

2.7 Flight at Aerodromes

2.7.1 Aerodromes not having Air Traffic Control units

(1) An aircraft shall not fly within an aerodrome traffic zone of an aerodrome which does not have an air traffic control unit, except for the purpose of taking off, landing or observing the signals in the signal area with a view to landing; and the aircraft flying in such a zone for the purpose of observing the signals shall remain clear of cloud and at least 500 feet above the level of the aerodrome.

(2) The commander of an aircraft flying in such a zone or moving on such an aerodrome shall:

- (a) Conform to the pattern of traffic formed by other aircraft, or keep clear of the airspace in which the pattern is formed;

- (b) Make all turns to the left unless ground signals indicate otherwise; and
- (c) Take-off and land in the direction indicated by the ground signals or, if no such are displayed, into the wind, unless good aviation practice demands otherwise.

2.7.2 **Aerodromes having Air Traffic Control units**

The commander of an aircraft flying in the aerodrome traffic zone of an aerodrome having an air traffic control unit or moving on the manoeuvring area of such an aerodrome shall:

- (a) maintain a continuous watch on the appropriate radio frequency notified for the control communication at the aerodrome; and
- (b) not taxi, take-off or land without permission of the air traffic control unit.

2.7.3 **In VMC:**

All aerodrome traffic are under the control of Aerodrome Control Unit. Nevertheless, the onus of avoiding collisions rests with the pilot.

2.7.4 **In IMC**

Control of aircraft on the runway in use and in the air rests with the Approach Control Unit.

2.7.5 **Combined Services**

At some controlled aerodromes, the functions of the aerodrome and approach control may be performed by a single unit employing one set of radio communications channels.

2.8 Aerodrome Control Service

2.8.1 Pre-taxi information

Prior to taxiing for take-off, aircraft shall be advised of the following information, in the order listed, except for elements which it is known to have already received.

- (i) The runway to be used.
- (ii) The current surface wind direction and speed
- (iii) The QNH altimeter setting
- (iv) The air temperatures to be used in the case of turbine engine aircraft.
- (v) The current visibility in direction of take-off if less than 10 km
- (vi) The correct time.

2.8.2 Prior to take-off

The aircraft shall be advised of:

- (i) Any significant changes in the surface wind velocity, air temperature and visibility.
- (ii) Any significant changes in meteorological conditions in the take-off and climb out area.

2.8.3 Prior to entering traffic circuit

Prior to entering the traffic circuit, an aircraft shall be provided with the following elements of information in the order listed:

- (i) The runway to be used
- (ii) The mean surface wind velocity and significant variations
- (iii) The QNH altimeter setting

2.9 Basic altimeter setting procedures

2.9.1 Take-off and Climb

A QNH altimeter setting will be made available in the routine take-off instructions. Vertical distance during the climb will be by reference to altitudes until the transition altitude is reached.

2.9.2 **En-route**

Aircraft en-route will be flown at flight levels at all times during an IFR flight. The levels flown must conform to the semi-circular rule.

2.9.3 **Approach and Landing**

A QNH altimeter setting and a transition level number will be made available to aircraft in the routine approach and landing instructions. The vertical displacement of aircraft on initial approach above the transition level will be controlled by reference to flight levels.

On penetrating the transition level in the course of descending, the vertical displacement of the aircraft will be controlled by reference to altitudes.

2.10 **Essential information on aerodrome**

Essential information on aerodrome conditions shall be given to every aircraft. At certain aerodromes this information together with meteorological data may be provided in the form of automated broadcasts through an automatic terminal information service (ATIS). The information shall include information relating to the following:

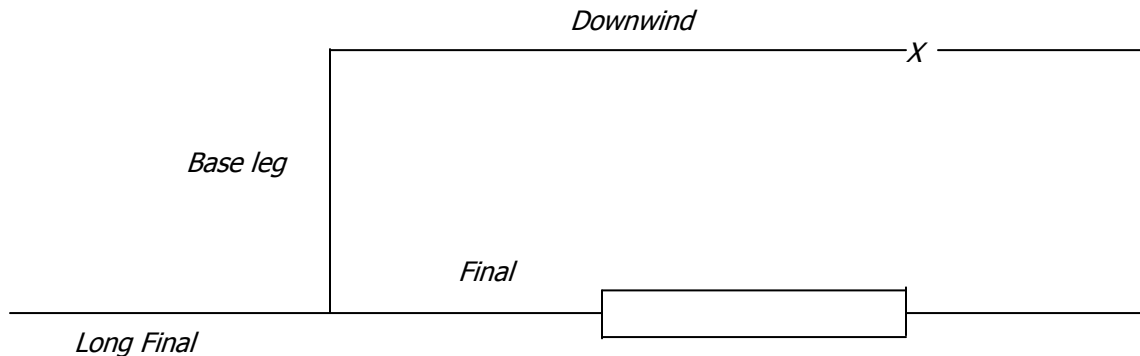
- (i) Construction or maintenance work on or adjacent to the maneuvering area
- (ii) Rough portions of any part of the maneuvering area
- (iii) Presence of water on runways and taxiways
- (iv) Parked aircraft or other objects on or immediately adjacent to taxiways.
- (v) Presence of other temporary hazards, such as binds on the ground or in the air.
- (vi) Failure of irregular operation of paid or all of the aerodrome lighting system.
- (vii) Any other pertinent information.

2.11 **Visual Circuit Reporting Procedure**

Rules regarding operation of aircraft at aerodromes have been stated in paragraph 2.7 above.

Paragraph 2.7.1 (2) (b) requires that commanders of aircraft make all turns to the left when approaching to land or after take-off unless ground signals indicate otherwise. This is the procedure for standard circuit.

In order that maximum use may be made of aerodrome for the purpose of landings and take-offs, it is essential that pilots accurately report their positions in the circuit when so requested. The Positions in which the various reports should be made are:



- (i) Downwind: when abeam the upwind end the runway
- (ii) Base leg: immediately on completion of turn on to base leg.
- (iii) Final: After the completion of the turn on to final approach when not more than 4 NM from the approach end of the runway.
- (iv) Long Final: Aircraft flying a final approach of a greater length than 4NM are to report "Long Final" when beyond that range and "Final" when a range of 4 NM is reached. (Long Final starts at 8 NM)

2.12 Use of Runways

2.12.1 Normally, only one aircraft will be permitted on the runway in use at any one time, however

2.12.2 An aircraft may be allowed to land on a runway before the preceding aircraft has cleared it if:

- (i) The length of the runway permits;
- (ii) It is during daylight hours; and
- (iii) The following aircraft will be able to see the preceding one clearly and continuously until it has turned off the runway.

The Aerodrome Control Unit will have warned the pilot of the aircraft ahead by giving the instruction "Land after..... (aircraft type)" instead of the usual "Cleared to land."

2.12.3 A departure aircraft may be cleared for take-off, or, an arriving aircraft may be cleared to land, when the preceding departing aircraft:

- (i) is airborne and has retracted its undercarriage; or

- (ii) (In the case of a fixed undercarriage aircraft), is airborne and has cleared the upwind end of the runway in use.

2.12.4 The pilot has the final decision as to which runway he uses.

2.13 **Entering a Control Zone or Control Area**

RTF Requests to enter a control zone or control area should be made when at least 10 minutes flying time from the boundary, giving the following information:

- (i) call sign or aircraft identification
- (ii) Type of aircraft
- (iii) Position, level and flight conditions
- (iv) Estimated time of arrival
- (v) True Air Speed

If aerodrome of departure is less than 10 minutes flying time from the boundary, pilots should request clearance before departure.

2.14 **Special Application of IFR**

When observed cloud base at an aerodrome within a control zone is more than 4 oktas below 1,500 ft or visibility is less than 8 km, all aircraft must obtain an ATC clearance or special VFR authorization before undertaking a flight within the control zone.

The following details shall be passed to an ATCU by an aircraft in flight when requesting a Special VFR clearance:

- (a) Call sign of the aircraft
- (b) Type of aircraft
- (c) Intentions
- (d) ETA at the point

2.15 **Crossing Airways**

Pilots of powered aircraft who wish to cross an airway in VMC by day may do so without compliance with full IFR requirements, provided that clearance is obtained from the appropriate ATS unit at least 10 minutes flying time from the intended crossing point.

Requests for crossing should include:

- (a) Aircraft identification
- (b) Aircraft type
- (c) True track
- (d) Place and estimated time of crossing
- (e) Desired crossing level
- (f) Groundspeed

2.16 **Pre-flight action**

- (1) A pilot-in-command shall, before commencing a flight, shall be familiar with all available information appropriate to the intended operation.
- (2) Pre-flight action by a pilot-in-command, for a flight away from the vicinity of the place of departure, and for every flight under the instrument flight rules, shall include-
 - (a) a careful study of available current weather reports and forecasts taking into consideration fuel requirements; and
 - (b) an alternative course of action if the flight cannot be completed as planned.

2.17 **Flight plan**

Except as authorized by the Authority a person shall not commence a flight if he has not filed a flight plan.

2.18 **Submission of a flight plan**

- (1) Information relating to an intended flight or portion of a flight, to be provided to air traffic services units, shall be in the form of a flight plan.

- (2) A pilot-in-command shall, prior to operating one of the following flights, must file a flight plan:
- (a) any flight, or portion thereof, to be provided with air traffic control service;
 - (b) any instrument flight rules flight within advisory airspace;
 - (c) any flight within or into designated areas, or along designated routes, when so required by the appropriate air traffic services authority to facilitate the provision of flight information, alerting and search and rescue services;
 - (d) any flight within or into designated areas, or along designated routes, when so required by the appropriate air traffic services authority to facilitate co-ordination with appropriate military units or with air traffic control units in adjacent states in order to avoid the possible need for interception for the purpose of identification;
 - (e) any flight across international borders; and
 - (f) any flight departing from an aerodrome manned by the Authority.
- (3) A pilot-in-command shall submit a flight plan before departure to the appropriate air traffic services reporting office or during flight (airborne flight plan), transmit to the appropriate air traffic services unit or air-ground control radio station, unless arrangements have been made for submission of a repetitive flight plan.
- (4) Unless otherwise prescribed by the Authority, a pilot-in-command shall submit a flight plan to the appropriate air traffic services unit-
- (a) at least sixty minutes before departure and shall be valid for sixty minutes for instrument flight rules flights or one hundred and twenty minutes for visual flight rules flights; or
 - (b) if submitted during flight, at a time which shall ensure its receipt by the appropriate air traffic control unit at least ten minutes before the aircraft is estimated to reach the:
 - (i) intended point of entry into a control area or advisory airspace; or
 - (ii) point of crossing an airway or advisory route.
- (5) Where a Through Flight Plan, containing such particulars as may be notified is submitted to and accepted by an air traffic services unit in

respect of a flight through a number of intermediate aerodromes, this regulation shall be deemed to have been satisfied in respect of each sector of the flight.

- (6) An air traffic control unit may exempt the pilot-in-command from the requirements of this regulation in respect of an intended flight which is to be made in a notified local flying area and in which the aircraft will return to the aerodrome of departure without making an intermediate landing.
- (7) In order to comply with the instrument flight rules, before an aircraft either takes off from a point within any controlled airspace, or enters any controlled airspace, or in other circumstances prescribed for this purpose, the pilot-in-command shall cause a flight plan to be communicated to the appropriate air traffic control unit and shall obtain an air traffic control clearance based on such flight plan.
- (8) The pilot-in-command after he has flown in controlled airspace shall, unless he has requested the appropriate air traffic control unit to cancel his flight plan, forthwith inform that unit when the aircraft lands within or leaves that controlled airspace.

CHAPTER 3

AERONAUTICAL PUBLICATIONS

3.1 Integrated Aeronautical Information Package

The aeronautical information is provided in the form of an Integrated Aeronautical Information Package. The package consists of:

- (i) Aeronautical Information Publication (A.I.P)
- (ii) Amendment Service to the AIP (A.I.P. AMDT)
- (iii) Supplement to the A.I.P (AIP SUP)
- (iv) NOTAM and Pre-Flight Information Bulletin (P.I.B)
- (v) Aeronautical Information Circulars (A.I.C)
- (vi) Checklist and Summaries.

3.2 Aeronautical Information Publication (A.I.P)

3.2.1 This is the basic aviation document intended primarily to satisfy international requirements for exchange of permanent aeronautical information and long duration temporary changes essential for air navigation.

3.2.2 Amendment service to the AIP (AIP AMDT)

Two types of AIP AMDT are produced.

- (a) Regular AIP Amendment (AIP AMDT) are issued twice in a year. The publication dates are on the first days of January and June every year and are identified by a light blue cover sheet.
- (b) AIRAC AIP Amendment (AIRAC AIP AMDT), issued in accordance with the AIRAC system and identified by a pink cover sheet and acronym AIRAC. It incorporates operationally significant permanent changes into the AIP on the indicated AIRAC effective date.

3.2.2 Supplement to The A.I.P (AIP SUP)

These comprise of temporary changes of long duration (3 months and longer) and information of short duration which consists of extensive and/ or graphics supplementing the permanent information in the AIP.

3.3 **NOTAM and Pre-Flight Information Bulletins (PIB)**

3.3.1 **NOTAM**

NOTAM contain information concerning the establishment, condition or change in aeronautical facility, service, procedure or hazard, the timely knowledge of which is essential for personnel concerned with flight operations.

NOTAM are originated and issued for the Nairobi FIR and are distributed in three series identified by the letters A, B and C.

Series A National and international distribution.

Series B International distribution to adjacent F.I.R. only.

Series C National distribution only.

Checklist of valid NOTAM are issued monthly via AFTN.

3.3.2 **Pre-Flight Information Bulletins (PIB)**

These are a recapitulation of current NOTAM and other information of urgent character for the operator/flight crew. They are available at aerodrome AIS Units.

3.4 **Aeronautical Information Circulars (AIC)**

These contain:

- (a) Information on long term forecast of any major change in legislation, regulations, explanatory or advisory nature liable to effect flight safety,
- (b) Information or notification of an explanatory or advisory nature liable to affect flight safety, and
- (c) Information or notification of an explanatory or advisory nature concerning technical, legislative or purely administrative matters.

Checklist of AIC currently in force is issued once a year

3.5 **AIRAC System**

These are issued on predetermined dates known as AIRAC System.

They consist of operationally significant changes requiring amendment of charts, route manuals, etc.

The System consists of effective dates on 28–day cycles and the information is issued so that it reaches the user:

- (i) 28 days Before the effective date, or
- (ii) 56 days before the effective date for major changes

3.6 **Pre-Flight Information Service at Aerodromes**

Pre-flight information is available at aerodromes as follow:

<u>Aerodromes</u>	<u>Briefing coverage</u>
1. NAIROBI/JKIA	All states within ICAO,AFI, EUR, MID, NAT and SAM Asia and Pacific Regions
2. MOMBASA/MOI	All states within ICAO, AFI, EUR, MID, Asia & Pacific Regions
3. ELDORET INT.	All states within ICAO, AFI, EUR, MID, Asia & Pacific Regions
4. WILSON	States within ICAO, AFI, Asia & Pacific Regions
5. MALINDI	} Adjacent F.I.R.s
6. KISUMU	
7. LOKICHOGGIO	
8. WAJIR	

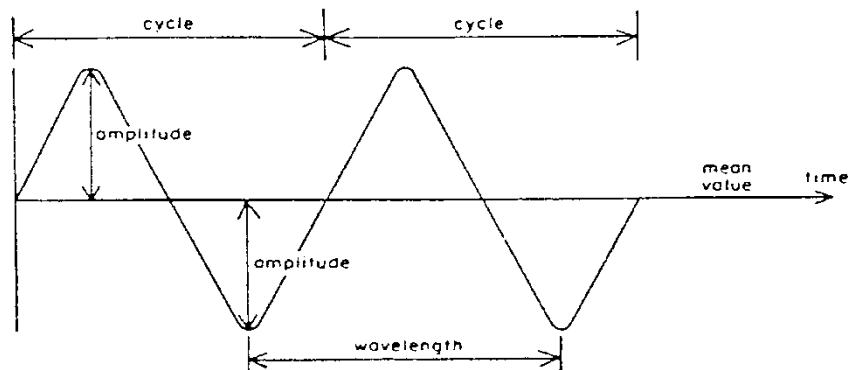
CHAPTER 4

RADIO WAVES

4.1 Electromagnetic Radio Waves

If an alternating current is fed to an aerial some of the power will be radiated outwards from the aerial in the form of ELECTROMAGNETIC RADIO WAVES.

4.2 Radio Wave Terminology



Cycle = one complete series of values

Phase = point reached by the wave front at any instant during the formation of a 360° cycle

Wavelength = distance traveled by the wave energy in the direction of propagation during one cycle; denoted by the Greek letter, λ .

Amplitude = The maximum displacement from the mean value

Frequency = Number of cycles which pass a fixed point in one second.

Hertz (Hz.) : One Hertz is one cycle per second.

4.3 Speed of Propagation

Radio waves travel approximately at the speed of light:-

162 000 Nautical Miles per Second	300 000 000 Metres per Second
186 000 Statute Miles per Second	300 000 Kilometres per Second

4.4 Frequencies

Since frequencies in the radio spectrum are measured in many thousands of Hz. The following prefixes are used:

1 KHz. (Kilo Hertz)	=	1,000 Hz.
1 MHz. (Mega Hertz)	=	1,000 KHz or 1,000,000 Hz.
1 GHz. (Giga Hertz)	=	1,000 MHz or 1,000,000,000 Hz.

4.5 Relationship between Wavelength and Frequency

The mathematical relationship between frequency and wavelength is:

$$\lambda = c/f \quad \text{where: } \begin{array}{l} c = \text{velocity of propagation (300 X } 10^6 \text{ m/s)} \\ f = \text{frequency (Hz)} \\ \lambda = \text{wavelength (m)} \end{array}$$

4.6 Frequency Bands

Waves used in aeronautical radio application vary in length from 1 mm to 100 km, and in frequency from 3 KHz to 300 GHz.

Different frequencies have different characteristics and are used for varying applications.

Radio waves are grouped into frequency bands as follows:

100km	10km	1km	100M	10M	1M	10Cm	1Cm	1Mm
VLF	LF	MF	HF	VHF	UHF	SHF	EHF	
3KHz	30KHz	300KHz	3MHz	30MHz	300MHz	3GHz	30GHz	300GHz

4.7 Frequency Wave Transmission

- (a) VLF, LF and MF
 - Are low frequency waves
 - They travel over the ground (ground waves) and tend to lose energy with time.
 - Some waves are transmitted upwards.
- (b) HF
 - They travel upwards and are reflected back to the ground by the ionosphere (sky waves).
 - This gives them a greater range than ground waves
- (c) VHF, UHF, SHF and EHF
 - These are very high frequency waves. They travel on the line of sight.

4.8 Propagation of Radio Waves

4.8.1 For convenience radio wave propagation is classified as:

- (a) Ground waves,
- (b) Sky waves, and
- (c) Direct waves

4.8.2 Ground Waves

Transmitted energy leaving the transmitter travels close to the ground following the curvature of the earth using a property known as diffraction. The amount of diffraction is depended on the wavelength and decreases steadily as wavelength is decreased. Ground waves make it possible to receive signals well beyond the visible horizon. Commonly used frequencies are VLF, LF and MF. HF is also used to some extent.

4.8.3 Sky Waves

In sky waves propagation the transmitted waves are returned to earth by refraction in the ionosphere. Ionosphere is that part of the outer atmosphere where solar (and cosmic) radiation removes electrons from the gas molecules and atoms resulting in positive ions and negative free electrons. Since the sun's energy is the main source of the ionization, densities, thicknesses and heights of the ionosphere differ between day and night. HF is the frequency commonly used in sky waves propagation.

4.8.4 Direct Waves

Direct waves follow a path similar to the line of sight, and so their surface range is limited by the curvature of the earth. Direct waves are present at all propagation frequencies, but the term is more used with VHF and above because at these frequencies the ground wave is very short and there is no sky wave refraction from the ionosphere.

Maximum theoretical range (MTR) for VHF and UHF direct communications can be calculated by:

$$\text{MTR} = 1.25 (\sqrt{\text{Height of Transmitter}} + \sqrt{\text{Height of Receiver Aerial}})$$

CHAPTER 5

AERONAUTICAL MOBILE SERVICE

5.1 Carriage of Radio Equipment

5.1.1 A person shall not operate an aircraft unless it is equipped with radio equipment-

- (a) that complies with the law of the State of registry;
- (b) required for the kind of operation being conducted; and
- (c) capable of receiving meteorological information at any time during the flight.

5.1.2 All aircraft operated under visual flight rules or instrument flight rules shall be equipped with radio communication equipment capable of conducting two-way communication with those aeronautical stations and on the frequencies prescribed by the Authority.

5.2 Definitions

5.2.1 **Aeronautical Mobile Service (AMS):** This is a mobile service between aeronautical stations and aircraft stations, or between aircraft stations.

NB: Emergency position indicating radio beacon stations or survival craft stations may also participate in this service on designated distress and emergency frequencies.

5.2.2 **Aeronautical Station:** is a land station in the aeronautical mobile service. In certain instances, it may be located on board a ship or on a platform at sea.

5.2.3 **Aeronautical Fixed Service:** A telecommunication service between specified fixed points provided primarily for the safety of air navigation and or the regular efficient and economical operation of air services.

5.2.4 **Aeronautical Fixed Station:** A station in the aeronautical fixed service.

5.2.5 **Air-ground communication:** Two way communication between aircraft and stations or locations on the surface of the earth.

5.2.6 **Air to ground communication:** One way communication from aircraft to stations or locations on the surface of the earth.

5.3 **Discipline in the AMS**

Highest standards of discipline shall be observed at all times in all communications in the AMS as follow:

- (a) Standard radiotelephony phraseology shall be used in all situations for which it is specified
- (b) Transmission of messages other than the authorized categories of message shall be avoided.
- (c) Where it is necessary for an aircraft station to send signals for testing or adjustment of radio apparatus the concord of the aeronautical station must be obtained and such transmission shall be kept to a minimum.
- (d) When it is necessary for a station to make test signals, such signals shall not continue for more than 10 seconds and shall be composed of spoken numerical (*ONE, TWO, THREE, ETC*). In radiotelephony, followed by the radio call sign of the station transmitting.
- (e) Except as otherwise provided, the responsibility of establishing communication shall rest with the station having traffic to transmit.
- (f) After a call has been made to an aeronautical station, a period of 10 seconds should elapse before a second call is made.
- (g) When an aeronautical station is called simultaneously by several aircraft stations, the aeronautical station shall decide the order in which aircraft shall communicate.
- (h) For any communication between aircraft stations on an ATS frequency, prior permission of the aeronautical station shall be obtained. Such request is not required for brief exchanges.

5.4 **Categories of Messages**

The categories of messages handled by the aeronautical mobile service and the order of priority shall be in accordance with the following:

- 5.4.1 Message categories & order of priority RTF signal
- | | | |
|-----|---|------------|
| (a) | Distress calls, distress messages & distress traffic | MAYDAY X 3 |
| (b) | Urgency messages, including messages proceeded by the medical transports signal | PAN X 3 |
| (c) | Communication s relating to direction finding | - |
| (d) | Flight safety messages | - |
| (e) | Meteorological messages | - |
| (f) | Flight regularity messages | - |
- NB:** NOTAM may qualify for any of the categories or priorities (c) to (f) depending on the contents of the NOTAM and its importance to the aircraft concerned.
- 5.4.2 Flight safety messages:
- Flight safety messages shall comprise the following
- (a) Movement and control messages.
 - (b) Messages originated by aircraft operating agency or by aircraft of immediate concern to an aircraft in flight.
 - (c) Meteorological advice of immediate concern to an aircraft in flight or about to depart.
 - (d) Other messages concerning aircraft in flight or about to depart.
- 5.4.3 Movement and Control Messages
- Movement and control messages shall comprise of:
- (a) Flight plan messages
 - (b) Amendment & co-ordination messages
 - (c) Cancellation messages
 - (d) Clearance messages
 - (e) Transfer of control messages
 - (f) Request messages
 - (d) Position report messages.
- 5.4.4 Meteorological Messages
- Meteorological messages shall comprise meteorological information to or from aircraft, other than those falling under flight safety, above.

5.4.5 Flight regularity Messages

The messages shall comprise the following:

- (i) Messages regarding the operation or maintenance of facilities
- (ii) Messages concerning the servicing of aircraft
- (iii) Instruction to aircraft operating agency concerning changes in requirements for passengers and crew
- (iv) Messages concerning non- routine landings to be made
- (v) Messages concerning aircraft parts and materials urgently required
- (vi) Messages concerning changes in aircraft operating schedules

- NB: 1. ATS units may handle these messages using RTF provided their primary role shall not be interfered with, and no other channels are available.
2. Messages having same priority should, in general, be transmitted in the order in which they are received for transmission.

CHAPTER RADIOTELEPHONY PROCEDURES

6.1 Standard Radiotelephony Phraseologies

Standard radiotelephony phraseologies should always be used when they are applicable. They have been devised to give both speed and intelligibility and because they are familiar to all users, will be more readily understood in difficult operating conditions than non-standard phraseologies

6.2 Language to be used

- (a) Air-ground RTF communication shall, in general, be conducted in the language normally used by the ground station.
- (b) In Kenya, the language is ENGLISH.

6.3 Word Spelling

When proper names, service abbreviations and words of which the spelling is doubtful are spelled out, the ICAO phonetic alphabet shall be used.

See phonetic radiotelephony alphabets are at APPENDIX 1.

6.4 Transmission of Numbers

6.4.1 Number and numeral elements shall be pronounced as follows:

<u>Numeral</u>	<u>Pronunciation</u>		
0	ZE-RO	6	SIX
1	WUN	7	SEV en
2	TOO	8	AIT
3	TREE	9	NIN er
4	FOW er	Decimal	DAY-SEE-MAL
5	FIFE	Thousand	TOU-SAND

Note: The syllables printed in capital letters in the above list are to be stressed

- 6.4.2 (a) All numbers except whole thousands shall be transmitted by pronouncing each digit separately.

- (b) Whole thousands are transmitted by pronouncing each digit in the number of thousands followed by the word TOU SAND.

Example:

<u>Number</u>	<u>Transmitted as</u>
10	WUN ZERO
75	SEVen FIFE
100	WUN ZERO ZERO
583	FIFE AIT TREE
5000	FIFE TOUSAND
25,000	TOO FIFE TOUSAND
25,500	TOO FIFE FIFE ZERO ZERO
38,143	TREE AIT WUN FOWER.TREE

6.4.3 Numbers Containing Decimals

Numbers containing decimal points shall be transmitted with the decimal point in appropriate sequence being indicated by the word DECIMAL.

Example:

<u>Number</u>	<u>Transmitted as</u>
118.1	WUN WUN AIT DAY-SEE-MAL WUN
100.3	WUN ZERO ZERO DAY-SEE-MAL TREE
3814.9	TREE AIT WUN FOWER TREE DAY-SEE-MAL NINer

6.4.5 VHF Frequencies

For identification of VHF frequencies, not more than two significant digits after the decimal point are used. A single zero is to be considered significant.

Example:

<u>VHF Frequency</u>	<u>Transmitted as</u>
118.0	WUN WUN AIT DAY-SEE-MAL ZERO
118.1	WUN WUN AIT DAY-SEE-MAL WUN
118.125	WUN WUN AIT DAY-SEE-MAL WUN TOO
118.150	WUN WUN AIT DAY-SEE-MAL WUN FIFE

6.4.6 **Transmission of time**

When transmitting time, only the minutes of the hour should be pronounced separately. However, the hours should be included when possibility of confusion is likely.

Example:

<u>Time</u>	<u>Transmitted as</u>
0920 (9:20 A.M)	TOO ZERO or ZERO NINer TOO ZERO
1643 (4:3 A.M)	FOWer TREE or WUN SIX FOWer TREE

6.5 **Transmitting Technique**

6.5.1 Each written message should be read prior to commencement of transmission in order to eliminate unnecessary delays in communication

6.5.2 Air crew and ground personnel should:

- (i) Enunciate each word clearly and distinctly
- (ii) Maintain an even rate of speech, not exceeding 100 words per minute
- (iii) Maintain the speaking volume at constant level
- (iv) Be familiar with microphone operating techniques
- (v) Suspend speech temporarily if it becomes necessary to turn head away from the microphone.

6.6 **Use of Abbreviated Words**

Abbreviated words or phrases should be transmitted in the unabbreviated words except for those owing to frequent and common practice, are normally understood by aeronautical personnel.

6.7 **Phonetic Spelling**

To expedite communication, the use of phonetic spelling should be dispensed with, if there is no risk of affecting reception intelligibility of the message.

6.8 **Transmission of Long Messages**

Transmission of Long messages should be interrupted momentarily from time to time to permit:

- (i) The transmitting operator to confirm the frequency in use is clear; and if necessary.
- (ii) The receiver to request repetition of parts not received.

6.9 **Composition of Message**

Messages handled entirely by the AMS shall comprise the following parts in the order stated:

- (i) Call indicating the addressee and the originator
- (ii) Text

Example: (i) (Call): WILSON TOWER 5YBBC
(ii) (Text): REQUEST JOINING INSTRUCTION.

6.10 **Words and Phrases to be used in RTF**

See APPENDIX 2

6.11 **Radiotelephony call signs for aeronautical stations**

Aeronautical stations in the AMS shall be identified by:

- (i) The name of the locations, and
- (ii) The unit or service available.

Example: See APPENDIX 3

6.12 **Radiotelephony call signs for aircraft**

6.12.1 An aircraft radiotelephony call sign shall be one of the following types:

- Type (a):** Five character call sign corresponding to the registration marking of the aircraft e.g. 5Y-ASA
- Type (b):** Five character call sign as in (a) above preceded by the RTF designator of Operator e.g. ZEBRA 5Y-BSA.
- Type (c):** Five character call sign as in (a) above preceded by the type of aircraft e.g. BUFFALO 5X-BFL.
- Type (d):** RTF designator of the aircraft operating agency followed by the flight identification e.g. KQA 275
- Type (e):** The characters corresponding to the registration marking of the aircraft e.g. N3782B.

6.12.2 Abbreviated RTF calls signs of aircraft

Type (a): The first letter or figure and the last two letters of the call sign; e.g. "5-SA" (for 5Y-ASA)

Type (b): The RTF designator of the agency followed by the last two letters of the call sign; e.g. "ZEBRA-SA"

Type (c): The type of aircraft followed by the last two letters of the call sign; e.g. "BUFFALO-FL"

Type (d): (No abbreviation)

Type (e): The first character and the last three characters of the call sign e.g. "N-82B"

NB: It is recognized that certain aircraft not governed by the above types employ for specific state purposes a coded call sign, e.g. "HAWK 22", "HARAMBEE-1", etc.

6.13 RTF Procedures

- 6.13.1 (a) An aircraft shall not change type of RTF call sign during flight, except temporarily on instruction by ATC.
- (b) No transmission shall be directed to an aircraft during take-off, during last part of final approach or during landing roll, except for reasons of safety.
- (c) Full RTF call signs shall be used always when establishing communication.
- (d) Communication shall commence with a call and a reply when it is designed to establish contact.

6.13.2 Subsequent RTF Communications

- (a) Abbreviated RTF call signs shall be used only after satisfactory communication has been established and provided that no confusion is likely to arise. An aircraft station shall use its abbreviated call sign only after it has been addressed in this manner by the aeronautical station.
- (b) After contact has been established, continuous two-way communication shall be permitted without further identification or call until termination of the contact.

- (c) In order to avoid any possible confusion, when issuing ATC clearances and reading back such clearances, controllers and pilots shall always add the call sign of the aircraft.
- (d) Read back of Instructions: The following items shall always be read back in radiotelephony communications
 - 1. ATC route clearances
 - 2. Clearances and instructions to enter, land on, take-off on, hold short of cross or backtrack any runway
 - 3. Runway in use, altimeter settings, SSR Codes, level instructions, heading and speed instructions.

NB: Messages not requiring read back are acknowledged by the aircraft call sign only.

6.14 **Radio Test Procedures**

- (a) The form of radio test transmission should be as follows:
 - (i) The identification of the station being called
 - (ii) The aircraft identification
 - (iii) The words "RADIO CHECK"
 - (iv) The frequency being used.
- NB:** A radio testing message should not exceed 10 seconds.
- (b) The reply to a test transmission should be as follows:
 - (i) The identification of the aircraft (or aeronautical station)
 - (ii) The identification of the aeronautical station (or aircraft) replying
 - (iii) The information regarding the readability of the aircraft transmission.
- (c) When tests are made, the following readability scale should be used:

- 1 Unreadable**
- 2 Readable now and then**
- 3 Readable but with difficulty**
- 4 Readable**
- 5 Perfectly readable**

6.15 **Air/Ground Communications Failure**

(a) When an aircraft station fails to establish contact with the aeronautical station on designated frequency, it shall attempt to establish contact on another frequency appropriate to the route. If the attempt fails, the aircraft shall attempt to establish communication with other aircraft or aeronautical stations on frequencies appropriate to the route.

(b) If attempts in 8.1.1 above fail, the aircraft shall transmit its message twice on the designated frequencies, preceded by the phrase "TRANSMITTING BLIND" including the addresses for which the message is intended.

NB: Note the difference between transmitting blind and broadcasting: In blind transmission two-way communication cannot be established but it is believed that the called station is able to receive the transmission, whereas in a broadcast, the transmitted information is not addressed to a specific station or stations.

(c) When an aircraft is unable to establish communication due to air-bone equipment failure it should, when so equipped, selected the appropriate SSR code to indicate radio failure. (A/C 7600).

6.16 **Basic Procedures to be adopted in the event of two-way Radio Communication Failure**

(a) Continue the flight in accordance with the current flight plan to the holding point at the aerodrome of first intended landing.

(b) Arrange the flight to arrive over the holding point at, or as close as possible to the estimated time of arrival in accordance with current flight plan.

(c) Land within 30 minutes of the time descent should have been started. If unable to land within this time, but able to complete an approach and landing visually, do so. Otherwise leave the vicinity of the aerodrome and associated controlled airspace on notified routes and altitudes (where applicable). If no altitude or route is specified, fly at last assigned altitude or minimum sector altitude, whichever is the higher, and avoid areas of dense traffic, then either:

- (i) fly to an area where flight can be continued in VMC and land at a suitable aerodrome there, or
- (ii) select a suitable area in which to descend through cloud, fly visually to suitable aerodrome and land as soon as practicable.

(d) In the event of a missed approach:

- (i) land at the aerodrome of destination if this can be achieved within 30 minutes of the time descent from the holding point should have started or (if this is impossible)
- (ii) leave controlled airspace at once in accordance with notified procedures.

Note: The message "Delay not determined" will not be considered to be an EAT. Pilots whose radio fail after they have received this message should not attempt to land at their planned destination aerodrome but should fly to another aerodrome following laid down procedures, if any.

CHAPTER 7

COMMUNICATIONS SERVICES

7.1 Extent of the Nairobi FIR

- (a) Laterally the Nairobi FIR extends from 04° N 034° E, – Eastward along 04° N to the Kenya/Somalia International borders; then southwards along the Kenya/Somalia border to a point 02° S 042° E, then from 02° S 042° E eastwards to a point 02° S 044° E, then from 02° S 044° E southward along the 044° E to 0442° S 044° E, then from 0442° S 044° E westward to a point 0442° S 039° E; thence along Kenya/Tanzania and Kenya/Uganda international borders to 04° N 034° E, from ground level to Unlimited; and
- (b) From 0442° S 04° E –South to a point 083° S 04° E, then North Westward to a point 0442° S 04° E, from above FL245 to Unlimited.

7.2 FIR Communication Frequencies

ACC: VHF: South Sector: 120.5 MHz and 125.3 MHz.
North Sector: 121.3 MHz and 118.5 MHz.

ATIS: 126.2 MHz Transmissions at 1 minute intervals

Unmanned aerodromes: 118.0 MHz.

HF: (See paragraph 7.4 below)

7.3 Communication Watch/Hours

During flight:

- (a) Aircraft stations shall maintain watch as required by the appropriate authority and shall not cease watch, except for reasons of safety, without informing the aeronautical station concerned.
- (b) Aircraft on long over water flights, or on flights over designated areas over which the carriage of survival radio equipment or emergency location because aircraft is required, shall continuously monitor the VHF emergency frequency 121.5 MHz
- (c) Aircraft on flights other than those specified in (b) above should continuously monitor the VHF emergency 121.5 MHz, except for those periods when they are carrying out communication on other

VHF channels or when air-borne equipment limitations or cockpit duties do not permit simultaneous monitoring of two channels.

- (d) The VHF Area Cover Scheme Frequency 118.5MHz is liable to interruption. Pilots should if unable to contact Nairobi Area Control Centre on VHF RTF change to HF RTF.

7.4 **HF Radio Frequencies in NAIROBI FIR**

The following HF radio frequencies are available in the Nairobi FIR as indicated:

- (a) Primary frequencies:
 - (i) 11300 KHz (Day)
 - (ii) 5517 KHz (Night)
- (b) Secondary frequencies:
 - (i) 13288 KHz
 - (ii) 3467KHz
 - (iii) 17961 KHz
 - (iv) 8870 KHz
 - (v) 6574 KHz
 - (vi) 8879 KHz

NB: All the frequencies are monitored throughout the day (H24).

7.5 **Principle of HF Network Operation (HF Communication)**

- (a) Aeronautical stations of a radiotelephony network assist each other in order to provide the air-ground communications service required of the network by aircraft flying on air routes for which the network is responsible.
- (b) When the network comprises a large number of stations, network communications for flights on any individual route segments is provided by selected stations, termed "regular stations" for that segment.

- NB:
1. Selection of a regular station is undertaken by regional or local agreement between the states responsible for the network.
 2. In principle, the regular station will be that securing the locations immediately concerned with flights on that route segment.
 3. In selecting the regular station, account is taken of the propagation characteristics of the frequencies.

- (c) Each regular station is responsible for:
 - (i) Designating suitable primary and secondary frequencies for its communication with the aircraft
 - (ii) Receiving all position reports and handling other messages from and to aircraft essential to the safe conduct of flight.
 - (iii) The action required in case of failure of communications

7.6 **RTF Procedures at Unmanned Aerodromes**

7.6.1 All aircraft operating at, or into, aerodromes which no ATS is provided must broadcast their position and intentions on VHF frequency 118.0 MHz.

7.6.2 Broadcast at, or into unmanned aerodromes shall include:

- (a) Initial call when in-bound.
 - (i) All stations (name of aerodrome)x2
 - (ii) Call sign and type of aircraft
 - (iii) Direction from the aircraft is inbound
 - (iv) EET in minutes to a position in the circuit or overhead.
 - (v) Any other information.
 - (vi) Position at which the next call will be made.
- (b) In the circuit, overhead or on the maneuvering area:
 - (i) (i) and (ii) above
 - (ii) Present position.
 - (iii) Intention of person in command.
 - (iv) Position at which next call will be made.

NB: 1. At aerodrome where ATS is normally provided but is temporarily unavailable for any reason, aircraft should broadcast their intentions on the promulgated aerodrome frequency and NOT on 118.0 MHz.

2. To minimize interference with communications of aircraft operating into adjacent aerodromes, position reports and intentions should NOT be broadcast outside a radius of 10 nm from the aerodrome or at greater height than 3000 ft above ground level.

3. A broadcast is a transmission of information relating to air navigation that is not addressed to a specific station or stations.

CHAPTER 8

AIRCRAFT OPERATION

8.1 Air Reports

- (a) An Air Report is a report from an aircraft in flight prepared in conformity with requirements for position, operations and/ or meteorological reporting.
- (b) Aircraft shall make air reports, of which position report is part on every IFR flight.
- (c) Air Reports transmitted by RT shall be transmitted in the AIREP Forms.

8.2 Reporting Points

Air Reports shall be required:

- (a) As soon as possible after passing a designated compulsory reporting point; or
- (b) On routes not defined by designated significant points, position reports shall be made as soon as practicable after transfer of control has taken place and at hourly intervals thereafter.

8.3 Operation Normal Procedures

- (a) The procedure is applied with the object of facilitating SAR.
- (b) All aircraft required to make air reports shall, in addition, transmit and receive acknowledgement for at least one message in the time interval between any two reports that are more than 40 minutes apart. This message may be sent by RT in the form "Operations Normal."
- (c) VFR lights shall make RT contact with appropriate ATS Units on the en-route frequency in accordance with the following procedures:
 - (i) As soon as possible after departure.
 - (ii) When changing frequency
 - (iii) When destination is in sight

- (iv) On flights of sufficient duration, an "Operations Normal" call or position report shall be made at intervals of not more than one hour.
- (d) In the event of failure to establish contact, pilots should broadcast their reports.

8.5 **Air Report Format**

Air Reports shall be made in the following form:

- (a) SECTION I POSITION REPORT
 - (i) Identification of aircraft
 - (ii) Position
 - (iii) Time
 - (iv) Flight Level (or VFR)
 - (v) Estimated Time of next Position
- (b) SECTION II OPERATING INFORMATION
 - (i) Estimated time of arrival
 - (ii) Endurance
- (c) SECTION III METREOLOGICAL INFORMATION
 - (i) Air Temperature
 - (ii) Wind
 - (iii) Turbulence
 - (iv) Aircraft Icing
 - (v) Supplementary Information

- 8.6
 - 1. Section I: shall always be included in Air Reports
 - 2. Section II: shall be included when specified by ATS Unit or the Operator
 - 3. Section III: shall be made at specified meteorological reporting point. It shall also be included in all IFR flights with a scheduled duration exceeding two hours; except when such flights are within one hour's flight from destination station or 30 minutes' flight from the departure station.

- 8.7 Position reports made by VFR aircraft shall include Section I and item (i) of Section II, and shall include item (ii) of Section (ii) on first contact if no flight pan has been filed.

8.8 **Designated Meteorological Reporting Points**

- (a) The boundaries of the FIR.
- (b) At hourly intervals when flying over the Indian Ocean at a distance of 30 minutes' flying time or greater from the shore irrespective of the flight level.
- (c) At any time when the observed weather differs significantly from the forecast given.
- (d) Pilots are requested to pass to an appropriate ATS Unit observations on adverse weather on climb out or approach to land.

Air reports shall be addressed to the appropriate ATS unit except that the last position report before passing from one FIR to an adjacent FIR shall also be addressed to the ATS unit providing flight information service or ATC service within the FIR that the aircraft is entering.

8.9 **Special Air Reports**

Special Air Reports shall be made by all aircraft on all routes whenever:

- (a) Severe icing or turbulence is encountered; or
- (b) Other meteorological conditions are encountered which in the opinion of the pilot in command are likely to affect safety of other aircraft. Such air reports should be prefixed "AIREP SPECIAL"

8.10 **Additional Air Reports**

Additional air reports shall be made if a specific request is made before or during flight by a meteorological service for the flight. Such air reports should be prefixed "AIREP ADD".

CHAPTER 9

DISTRESS AND URGENCY TELEPHONY COMMUNICATIONS PROCEDURES

9.1. Definition

Distress and urgency traffic shall comprise all radiotelephony messages relative to the distress and urgency conditions respectively. The significance of these messages is to alert other stations the existence of the relevant emergency situation for it to be offered the due priority. Distress and urgency conditions are defined as:

- 9.1.1 **Distress:** This is a condition of being threatened by serious and/or imminent danger and requiring immediate assistance
- 9.1.2 **Urgency:** A condition concerning the safety of an aircraft or other vehicle, or some person on board or within sight, but which does not require immediate assistance.

The radiotelephony distress signal MAYDAY x3 and the radiotelephony urgency signal PAN X3. , shall be used at the commencement of the first distress and urgency communication respectively.

At the commencement of any subsequent communication in distress and urgency traffic, it shall be permissible to use the radiotelephony distress and urgency signals.

9.2 Radiotelephony Distress Communication

9.2.1 The distress message to be sent by an aircraft in distress shall:

- (a) Be on the air-ground frequency in use at the time;
- (b) Consist of as many as possible of the following information elements spoken distinctly and, if possible, in the following order:
 - (i) Name of the station addressed (time and circumstances permitting);
 - (ii) The identification of the aircraft;
 - (iii) The nature of the distress condition;
 - (iv) Intention of the person in command; and
 - (v) Present position.

Note: The station addressed will normally be that communicating with the aircraft or whose area of responsibility the aircraft is operating.

- (c) Telephone calls concerning distress incidents may be made from any telephone. They should be passed in the form EMERGENCY to

9.2.2 Distress Frequency

- (a) An aircraft making an emergency signal should address its first call to the ATS Unit with which it has already established communication or to one which is known to be within range.
- (b) Should an aircraft be unable to establish communication on the A/G Frequencies already listed, every effort should be made to make contact with an aircraft, land or ship station on either:
 - (i) 121.5 MHz (Emergency Locator Beacon –Aircraft: ELBA. This assists in locating an aircraft crash site.); or
 - (ii) 243 MHz (ELBA for Military); or
 - (iii) 500 KHz (International distress frequency); or
 - (iv) 2182 KHz (Maritime Mobile Distress Frequency); or
 - (v) Any other available frequency.

- NB: 1 There is more chance of signals being heard on 500 KHz during the international silence period at 15 to 18 minutes and 45 to 48 minutes past every hour.
2. The emergency frequency 121.5 MHz. may be used to practice emergency procedures including simulated urgency but shall NOT distress situations

9.2.3 Amateur Radio Stations

In the event of an aircraft having made a forced landing, in a remote region, further attempts may be made to communicate with amateur stations on either:

- (a) 3650 KHz (night); or
- (b) 7075 KHz (day); or
- (c) 14150 KHz (day and night)

9.2.4 **Action by station addressed or first station acknowledging the Distress Message**

The station addressed by aircraft in distress or first station acknowledge the distress message shall:

- (a) Immediately acknowledge the distress message.
- (b) Take control of the communication or specifically and clearly transfer that responsibility, advising the aircraft if a transfer is made.
- (c) Take immediate action to ensure all necessary information is made available as soon as possible to:
 - (i) The ATS Unit concerned
 - (ii) The aircraft operating agency concerned or its representatives
- (d) Warn other stations, as appropriate in order to prevent the transfer of traffic to the frequency of the distress communication.

9.2.5 **Imposition of Silence**

The station in distress or the station in control of distress traffic shall be permitted to impose silence, either on all stations of the mobile service in the area or any station which interferes with distress traffic. The instructions shall be addressed as follow:

- "- All stations" (or to one specific station)
- STOP TRANSMITTING
- MAYDAY."

9.2.6 **Action by other Aircraft**

- (a) Distress communications have absolute priority over all other communication, and stations aware of distress communications shall not transmit on the frequency concerned unless:
 - (i) The distress is cancelled or the distress traffic is terminated.
 - (ii) The distress traffic has been transferred to other frequencies.
 - (iii) The station controlling communication gives permission.
 - (iv) It has itself to render assistance.
- (b) Any station which has knowledge of distress traffic, and which cannot itself assist the station in distress, should nevertheless continue listening to such traffic until it is evident that assistance is being provided.

9.2.7 Termination of Distress Communications and Silence

- (a) When an aircraft is no longer in distress, it shall transmit a message canceling the condition.
- (b) When the station which has controlled the distress communication traffic becomes aware that the distress condition is ended, it shall take immediate action to ensure that the information is made available, to:
 - (i) The ATS unit concerned;
 - (ii) The aircraft operating agency concerned, or its representative, in accordance with pre-established arrangements
- (c) The distress communication and silence conditions shall be terminated by transmitting a message, including the words "DISTRESS TRAFFIC ENDED" on the frequency or frequencies being used for the distress traffic.

9.3 Radiotelephony Urgency Communications

9.3.1 The urgency message to be sent by an aircraft reporting an urgency condition shall:

- (a) Be on the air-ground frequency in use at the time;
- (b) Consist of as many as required of the following elements spoken distinctly and, if possible, in the following order:
 - (i) The name of the station addressed;
 - (ii) The identification of the aircraft;
 - (iii) The nature of the urgency condition;
 - (iv) Intention of the person in command; and
 - (v) Present position, level (i.e. flight level, altitude, etc. as appropriate) and heading.
 - (vi) Any other useful information.

Note: The station addressed will normally be that communicating with the aircraft or whose area of responsibility the aircraft is operating.

9.3.2 **Action by station addressed or first station acknowledging the urgency message**

The station addressed by aircraft reporting an urgency condition or first station acknowledging the urgency message, shall:

- (a) Acknowledge the urgency message.
- (b) Take immediate action to ensure all necessary information is made available as soon as possible to:
 - (i) The ATS Unit concerned
 - (ii) The aircraft operating agency concerned or its representatives in accordance with pre-established arrangements.
- (c) If necessary, exercise control of the communication.

9.3.3 **Action by all other stations**

The urgency communications have priority over all other communication except distress, and all stations shall take care not to interfere with the transmission of urgency traffic.

9.4 **Action by an aircraft used for medical transports**

9.4.1 For the purpose of announcing and identifying aircraft used for medical transports, a transmission of the radiotelephony urgency signal "PAN PAN PAN" shall be followed by the radiotelephony signal for medical transports "MEDICAL"; pronounced "MAY-DEE-CAL". The use of these signals indicates that the message which follows concerns a protected medical transport. The message shall convey the following data:

- (a) The call sign or other recognized means of identification of the medical transport;
- (b) Position of the medical transport;
- (c) Number and type of medical transports;
- (d) Intended route
- (e) Estimated time en-route and of departure and arrival, as appropriate; and
- (f) Any other information such as flight altitude, radio frequencies guarded, languages used, and SSR modes and codes.

9.4.2 **Action by station addressed or first station acknowledging the Distress Message**

The provisions of 9.2.4 and 9.3.3 shall apply as appropriate to stations receiving a medical transport report.

CHAPTER 10

SEARCH AND RESCUE (SAR)

10.1 Emergency Phases

For the purpose of SAR, aircraft emergencies are classified in the following three phases:

- (a) **Uncertainty Phase (INCERFA):** – a situation where in uncertainty exists as to the safety of an aircraft and its occupants.
- (b) **Alert Phase (ALERFA):** - A situation wherein apprehension exists as to the safety of an aircraft and its occupants.
- (c) **Distress Phase (DETRESFA):** - A situation wherein there is reasonable certainty that an aircraft and its occupants are threatened by grave danger and require immediate assistance.

10.2 Circumstances requiring declaration of SAR Phases

10.2.1 Uncertainty Phase (INCERFA)

- (a) No communication has been received from an aircraft within a period of thirty minutes after time a communication should have been received, or the time unsuccessful attempt to establish communication with such aircraft was first made, whichever is the earlier; or
- (b) An aircraft fails to arrive within thirty minutes of estimated time of arrival last notified to or estimated by air traffic services units, whichever is the later, except when no doubt exists as to the safety of aircraft and its occupants

10.2.2 Alert Phase (ALERFA)

- (a) Following the uncertainty phase, subsequent attempts to establish communication with the aircraft or inquiries or other relevant sources have failed to reveal any news of the aircraft; or when
- (b) An aircraft has been cleared to land and fails to land within five minutes of the estimated time of landing and communication has not been re-established with the aircraft; or when
- (c) Information has been received which indicates that the operating efficiency of the aircraft has been impaired, but not to the extent

that a forced landing is likely, except when evidence exist that would allay apprehension as to the safety of the aircraft and its occupants; or when

- (d) An aircraft is known or believed to be the subject of unlawful interference.

10.2.3 **Distress Phase (DETRESFA)**

- (a) Following the alert phase, further unsuccessful attempts to establish communication with the aircraft and more widespread unsuccessful inquiries point to the probability that the aircraft is in distress; or when
- (b) The fuel on board is considered to be exhausted, or to be insufficient to enable the aircraft to reach safety; or when
- (c) Information is received which indicates that the operating efficiency of the aircraft has been impaired to the extent a forced landing is likely; or when
- (d) Information is received or it is reasonably certain that the aircraft is about to make or has made a forced landing, except when there is reasonable certainty that the aircraft and its occupants are not threatened by grave and eminent danger and do not require immediate assistance.

10.3 **SAR Units**

Whenever SAR operations are required, a Rescue Coordination Centre (RCC) is established adjacent to the Nairobi Area Control Centre.

Type I Rescue Sub-Centres (RSCS) have been established in all ATC Units; while Type II Sub-Centres are to be found at all Provincial Police Headquarters

10.4 **Notification of Rescue Coordination Centres**

Notification shall contain as much of the following information as is available in the order listed:

- (a) INCERFA, ALERFA or DETRESFA as appropriate to the phase of the emergency;
- (b) Agency and person calling;
- (c) Nature of emergency;
- (d) Significant information from the flight plan;

- (e) Unit which made last contact, time and means used;
- (f) Last position report and how determined;
- (g) Colour and distinctive marks of aircraft;
- (h) Dangerous goods carried as cargo;
- (i) Any action taken by reporting office; and
- (j) Other pertinent remarks.

10.5 **Ground/Air Visual Codes for use by Survivors**

No.	Message	Code Symbol
1.	Require assistance	V
2.	Require medical assistance	X
3.	No or Negative	N
4.	Yes or Affirm	Y
5.	Proceeding in this direction	↑

10.6 **Facilities Available**

Facilities available for SAR may vary from time to time. An up-to-date file is kept in the Area Control Centre.

The following are the facilities of permanent nature:

- (a) **Kenya Aircraft:** Kenya Air Force, Police Air Wing, Kenya Airways, Charter Companies, Aero Club, Private Companies/Owners, Par-Rescue Units
- (b) **Ground Parties:** Kenya Army, National Youth Service, Provision Administration, Forest Rangers, Mountain Rescue, Kenya Police and Game Department
- (c) **Marine:** Kenya Navy, Kenya Ports Authority, Coastal Shipping, Police (Lake Victoria)
- (d) **Communication (Radio):** The SAR is responsible for SAR operations within Nairobi FIR.

10.7 **The International 406 MHz. Beacon Registration Database (IBRD)**

10.7.1 The system became effective on 16th January 2006. it is managed by the COSPAS-SARSAT Secretariat as a supporting element of the satellite alert system that detects and distributes signals transmitted by 406MHz. Emergency Locator Transmitters. Signals transmitted by the system are of digital data nature with a unique identification.

Note: The COSPAS-SARSAT System will cease processing of 121.5/243 MHz. signals with effect from February 1, 2009. All 121.5/243 MHz. Beacons should be upgraded to 406 MHz. before the end of this deadline.

10.7.2 **Services provided by COSPAS-SARSAT System**

- (i) Positions of activated ELT are relayed to SAR Units more quickly, more reliably and with greater accuracy.
- (ii) SAR Authorities can interrogate a registration database and retrieve the characteristics of a subject aircraft and contact details of the ELT owner.
- (iii) SAR alerts can be resolved as either genuine or false by a telephone call.

10.7.3 **Registration**

For the system to operate efficiently, owners should register their ELT and the SAR providers should have access to the registration database.

CHAPTER 11

DIRECTION FINDING

11.1 Working of Direction Finding Stations

Direction-finding stations work either singly or in groups under the direction of a main direction-finding station.

11.1.1 Direction-finding station working alone

Such a station can only determine the direction of an aircraft in relation to itself. It should give the following, as required:

- (a) QTE True bearing of the aircraft from the station
- (b) QUJ True heading to be steered by aircraft, with no wind, to head for the direction finding station.
- (c) QDR Magnetic bearing of aircraft, from the station.
- (d) QDM Magnetic heading to be steered by aircraft, with no wind, to make for the station

11.1.2 Direction-finding stations working as a network (Group)

When direction-finding stations work as a network, to determine the position of an aircraft, the bearing taken by each station is sent immediately to the station controlling the DF network to enable the position of the aircraft to be determined.

The station controlling the network gives aircraft's position in one of the following ways:

- (a) Position in relation to a point of reference or in latitude and longitude using the signal QTE.
- (b) True bearing of the aircraft in relation to the direction-finding station or other specified point, using the signal QTE and its distance for the direction-finding station or point, using the signal QGE.
- (c) Magnetic heading to steer with no wind, to make for direction finding station or other specified point using the signal QDM and the distance from the direction-finding station or point, using the signal QGE

11.2 Accuracy of DF Stations

The accuracy of direction-finding station in bearings and positions are classified as follows:

<u>Class</u>	<u>Bearings</u>	<u>Positions</u>
Class A	$\pm 2^0$	within 9.3 km (5nm)
Class B	$\pm 5^0$	within 37 km (20nm)
Class C	$\pm 10^0$	within 92 km (50nm)
Class D	< Class C	Less than Class

11.3 Service

- (a) Direction-finding stations shall have authority to refuse to give bearings, heading or positions when conditions are unsatisfactory or when bearings do not fall within the calibrated limits of the station, stating the reason at the time of refusal.
- (b) An aircraft requiring a series of bearings or headings, shall call the direction-finding station concerned, and request the service by the signal QDL followed by the appropriate Q signal.

Example:

Pilot:	"Wilson Homer 5Y-AAK request QDL ODM"
Homer	"5Y-AAK transmit for QDM"
Pilot:	"5YAAK (Hold (key) transmitter for seconds) 5YAAK"
Homer	"5YAAk, QDM 110° Alpha" (meaning: to reach station in zero wind conditions, steer magnetic heading 110° $\pm$ 2°).
Pilot:	"QDM 110° A 5YAAK"

- (c) A pilot transmission for direction finding should end the transmission with the aircraft call sign.

CHAPTER 12

METEOROLOGICAL SERVICES

12.1 Responsible Authority

Meteorological services for civil aviation in Kenya are provided by the Meteorological Department of the Ministry of Transport in accordance with provisions contained in ICAO documents:

- (i) Annex 3 – “Meteorological Services for International Air Navigation”
- (ii) Doc 7030 – “Regional Supplementary Procedures”

12.2 Area of Responsibility

Meteorological service is provided within the Nairobi FIR. Main analysis and forecast centre is located at the Meteorological Headquarters, Dagoretti Corner, Nairobi. It is responsible for the provision of synoptic analysis, area forecast products, warnings and guidance material to Aerodrome Meteorological Offices.

12.3 Observations and Reports

Meteorological observations and reports are made at:

- (i) JKIA ½ Hourly plus special observations
- (ii) Moi ½ Hourly plus special observations
- (iii) Eldoret ½ Hourly plus special observations
- (iv) Wilson ½ Hourly plus special observations
- (v) Malindi Hourly plus special observations
- (vi) Eastleigh Hourly plus special observations
- (vii) Kisumu Hourly plus special observations
- (viii) Lokichoggio Hourly plus special observations
- (ix) Wajir Hourly plus special observations

12.4 Types of Services

12.4.1 Personal briefing and consultation for flight crew are provided only at JKIA, Moi, Eldoret, Wilson, Kisumu, Lokichoggio, Malindi and Wajir. For all other aerodromes consultation is available by telephone.

12.4.2 (a) Limited flight documentation is normally provided for domestic flights.

- (b) For international flights, documentation comprises a significant weather chart, an upper wind and upper air temperature chart and the latest available aerodrome forecast for the destination and alternate aerodromes

12.4.3 For the planning of VFR flights departing from unmanned aerodromes, pilots can obtain weather reports and forecasts from relevant aerodrome units on notified telephone numbers.

12.4.4 Whenever possible, requests for weather forecasts should be made at least 6 hours before the flight time and should include:

- (a) The route and/or aerodromes for which the forecasts are required.
- (b) The period of flight for which the forecasts are required.

12.5 **Aircraft Weather Reporting Procedures**

Captains in command of aircraft are required to make weather observations and pass weather reports over routine channels to the meteorological office.

Aircraft observations during flight are recorded on the AIREP form. Such forms should be handed in at the destination aerodrome at debriefing. Weather reports should be in accordance with provisions in Chapter 11 of this document.

12.6 **Transmission of Weather Elements**

The elements of actual weather reports to aircraft will always be transmitted in the following order:

- (i) Station and time of report;
- (ii) Surface wind, direction and speed
- (iii) Horizontal visibility (or RVR reading when applicable)
- (iv) (Present weather if significant)
- (v) Cloud amount and base
- (vi) Temperature and Dew point
- (vii) QNH (or QFE on request)
- (viii) Supplementary information in plain language

Notes: (a) Unless special circumstances necessitates the following words will not be included in transmission of weather elements to aircraft: "Wind", "Visibility", "Weather conditions" and "Cloud".

- (b) **CAVOK:** In passing the weather elements, items iii, iv and v may be substituted with the word "CAVOK" to imply:
 - (i) Visibility more than 10 kilometres;
 - (ii) No cloud below 5,000 feet above ground level; and
 - (iii) No precipitation reaching the ground, no thunderstorm activity observed and no shallow fog exists.
- (c) Dimension units used relating to wind direction and speed, visibility and cloud base will be transmitted.
- (d) Type of cloud will be mentioned only in the case of Cumulonimbus
- (e) The aircraft station will acknowledge receipt of the weather elements, read back the QNH setting and terminating by its call sign. e.g. "Weather copied. QNH 1024. 5Y-BBC".

RADIOTELEPHONY SPELLING ALPHABET

<u>Letter</u>	<u>Identification</u>	<u>Spoken as</u>
A	ALPHA	AL - FAH
B	BRAVO	BRAH - VOH
C	CHARLIE	CHAR - LEE
D	DELTA	DELL - TAH
E	ECHO	ECK - OH
F	FOXTROT	FOKS – TROT
G	GOLF	GOLF
H	HOTEL	HOH – TELL
I	INDIA	IN – DEE – AH
J	JULIET	JEW LEE ETT
K	KILO	KEY LOW
M	MIKE	MIKE
N	NOVEMBER	NO VEM BER
O	OSCAR	OSS CAH
P	PAPA	PAH PAH
Q	QUEBEC	KEH BECK
R	ROMEO	ROW ME OH
S	SIERRA	SEE AIR RAH
T	TANGO	TANG GO
U	UNIFORM	YOU NEE FORM
V	VICTOR	VIK TAH
W	WHISKY	WISS KEY
X	X-RAY	ECKS RAY
Y	YANKEE	YANG KEY
Z	ZULU	ZOO LOO

*Note: Syllables to be stressed are in **BOLD** type*

WORDS AND PHRASES TO BE USED IN RTF

<u>Phrase</u>	<u>Meaning</u>
ACKNOWLEDGE	Let me know.
AFFIRM	Yes.
ALL STATIONS	All aircraft stations on this frequency.
APPROVED	Permission for proposed action granted.
BREAK	I hereby indicate the separation between portions of the message. (to be used where there is no clear distinction between the text and other portions of the message).
BREAK BREAK	I hereby indicate the separation between messages transmitted to the different aircraft in a very busy environment.
CANCEL	Annul the previously transmitted clearance.
CHECK	Examine a system or procedure. (No answer is expected).
CLEARED	Authorized to proceed under the conditions specified.
CONFIRM	'Have I correctly received the message?' or 'Did you correctly receive the message?'
CONTACT	Establish radio contact with
CORRECT	That is correct.
CORRECTION	An error has been made in this transmission (or message indicated). The correct version is
DISREGARD	Consider that transmission as not sent.
GO AHEAD	Proceed with your message.

HOW DO YOU READ?	What is the readability of my transmission?
I SAY AGAIN	I repeat for clarity or emphasis.
MONITOR	Listen out on (frequency).
NEGATIVE	'No.' or 'Permission not granted' or 'That is not correct'.
ORBIT	Make a 360° turn.
OUT	This exchange of transmission is ended and no response is expected.
OVER	My transmission is ended and I expect a response from you.
READ BACK	Repeat all, or the specified communications part of the message back to me exactly as received.
RECLEARED	A change has been made to your last clearance and this new clearance supersedes your previous clearance or part thereof.
REPORT	Pass the following information
REQUEST	I should like to know or wish to obtain
ROGER	I have received all your last transmission.
SAY AGAIN	Repeat all, or the following part of your last transmission.
SPEAK SLOWER	Reduce your rate of speech.
STANDBY	Wait and I will call you.
UNABLE COMPLY	Not able to carry out received instruction.
VACATE	Leave or depart from
VERIFY	Check and confirm with originator.
WILCO	Abbreviation for 'Will comply.' I understand your message and will comply with it.

- WORDS TWICE
- (a) As a request: Communication is difficult. Please send every word or group of words twice.
 - (b) As information: Since communication is difficult, every word in this message will be sent twice.

APPENDIX

3

RADIOTELEPHONY CALL SIGNS FOR AERONAUTICAL STATIONS

<u>Unit/Service available</u>	<u>Call sign suffix</u>
Area Control Centre	CONTROL
Approach Control	APPROACH
Approach Control Radar arrivals	ARRIVAL
Approach control Radar departures	DEPARTURE
Aerodrome Control	TOWER
Surface movement control	GROUND
Radar (in general)	RADAR
Precision approach radar	PRECISION
Direction finding station	HOMER
Flight information service	INFORMATION
Clearance delivery	DELIVERY
Apron Control	APRON
Company dispatch	DISPATCH
Aeronautical station	RADIO

RTF PHRASEOLOGY EXERCISE

Aircraft Position	Aircraft Transmissions	ATS Transmissions
At Parking Area	<i>Wilson Tower</i> x2, <i>5Y-KSF</i> Radio Check on 118.1- Over	<i>5-SF Wilson Tower</i> read you 5
Ready to Taxi	Tower, <i>5Y-KSF</i> - At Base 1 - Taxi clearance for <i>Busia</i> (or for any other purpose) - POB 2, END 0530 Over. Holding Point 07. QNH 1024. <i>5-SF</i>	<i>5-SF</i> taxi Holding Point Rwy 07. QNH 1024
At Holding Point	<i>5-SF</i> at Holding Point, ready for departure – Over. Holding Position. <i>5-SF</i> Cleared Take-off Rwy 07, Right Turn <i>5Y-KSF</i>	<i>5-SF</i> Hold Position. <i>5Y-KSF</i> Cleared for take-off Rwy 07, Right Turn Out.
After Take-off	Departure 15 Will report Zone Out. <i>5-SF</i> .	<i>5-SF</i> departure time 15. Report Control Zone Boundary – over.
At CTR Boundary	Tower, <i>5-SF</i> Control Zone Boundary outbound. Nairobi Control. 118.5, <i>5-SF</i> .	<i>5-SF</i> contact Nairobi Control on 118.5 – Over.
Just after transfer to 118.5 MHz.	Nairobi Control x 2, <i>5Y-KSF</i> – over. <i>5-SF</i> departed <i>Wilson 0715</i> , Checked Zone Boundary <i>0721</i> .	<i>5Y-KSF</i> Nairobi Control, go ahead.

	VFR, Estimating <i>Busia</i> 0910. Will report Operations Normal 0825. 5-SF. .	5-SF Roger. Report Operations Normal 0825.
At 0825	Nairobi Control, 5-SF on course to <i>Busia</i>, reporting OPS Normal WILCO. 5-SF	5-SF Roger. Next report <i>Busia</i> in sight.
At 0840	MAYDAY, MAYDAY, MAYDAY. -NAIROBI CONTROL - 5Y-KSF - Bird strike, Port engine problem. - Making a forced landing in an open field. - Position: 15 NM south of Nandi Hills.	MAYDAY, 5Y-SKF, Nairobi Control Roger, MAYDAY.
At 0845	- Nairobi Control,- 5Y-SKF 5Y-SKF - Overhead Nandi Hills airstrip - Engine problem cleared, continuing flight to <i>Busia</i> - Estimate <i>Busia</i> 0915. - DISTRESS TRAFFIC ENDED WILCO, 5-SF	5Y-SKF Go-ahead. 5Y-KSF Nairobi Control Roger. DISTRESS TRAFFIC ENDED. 0845. Report <i>Busia</i> in sight
At 0900 and (Destination) in sight	Nairobi Contol, 5-SF <i>Busia</i> in sight, landing in 10 minutes. Changing to 118.0,5-SF.	5-SF Roger, Change to 118.0
After change to 118.0 MHz.	All stations at <i>Busia</i> x2 - 5Y-KSF, PA34, - Inbound from <i>the South-east</i>. - Estimating <i>Busia</i> 0915	

	-For landing - Will report again overhead <i>Busia</i>.	
When overhead the airfield (500 ft. above AD traffic Circuit)	All stations at <i>Busia</i> x2 - 5Y-KSF, PA34 - Overhead <i>Busia</i>, - Will report joining Left Downwind Rwy 09 for landing.	
In the Circuit	- All station at <i>Busia</i> x2 - 5Y-KSF PA34 on Left Downwind Rwy 09 for landing. - All station at <i>Busia</i> x2 - 5Y-KSF PA34 on Left Base Rwy 09 for Landing. - All station at <i>Busia</i> x2 - 5Y-KSF PA34 Final Rwy 09 Landing.	
On ground, vacating the runway.	- All station at <i>Busia</i> x2 - 5Y-KSF just landed, vacating Rwy 09.	
On ground, departing from unmanned aerodrome.	All stations at <i>Busia</i> x2 5Y-KSF, PA34, backtracking Rwy 09 for departure. Will report departing.	
Departing	All stations at <i>Busia</i> x2 - 5Y-KSF, PA34, - On Rwy 09 Commencing Take-off.	
Upon leaving the aerodrome traffic zone.	Nairobi Control x 2, 5Y-KSF – Over. - 5Y-KSF Departed <i>Busia</i> 1035, for <i>Wilson Airport</i> - VFR	5Y-KSF Nairobi Control, go ahead.

	- Estimating <i>Wilson 1215</i>. - 4 POB, <i>Endurance 0300 Hours</i> Will report Operations Normal 1145 5-SF.	5-SF, Nairobi Control, next Report OPS Normal 1145.
At 1130.	Nairobi Control, 5Y-KSF - Passing overhead Kericho - True Track 180°. - Request permission to cross Airway A609 South of Kericho at 1140 - VFR FL105 - G/S 145 KT Over.	5-SF crossing A609 South of Kericho approved. Report crossing.
At 1140.	Control, 5Y-KSF crossing Airway A609 1140, VFR FL105.	5-SF, Roger. Report leaving the airway.
At 1155.	Control, 5-SF leaving Airway A609. Over. Will report Wilson in sight.5-SF.	5-SF Roger, Next report <i>Wilson</i> in sight.
At 1204.	Control, 5-SF - Approaching Control Zone Boundary - <i>Wilson</i> in sight. <i>Wilson Tower 118.1</i>	5-SF Roger. Contact <i>Wilson Tower 118.1</i> . Over.
At 1205.	<i>Wilson Tower x2, 5Y-KSF</i>. Over. - 5-SF from <i>Busia</i> - VFR - Crossing zone boundary from the Southwest. - Estimate <i>Wilson 1215</i> - Request joining instruction for	5-SF, <i>Wilson Tower</i> , go ahead.

	landing. QNH 1021. Join Left Downwind RWY 25; 5-SF.	5-SF Roger. Join Left Downwind Rwy 25. QNH 1021 Over.
Joining Downwind	Tower –5-SF Joining Left Downwind Rwy25. Over 5-SF WILCO. Number Two. Caravan in sight	5-SF Number Two behind the Caravan on final Rwy 25. Report Final.
Turning Final	Tower, 5-SF Final Rwy 25. Caravan in sight To land after the Caravan 5-SF	5-SF Land Rwy 25after the Caravan
Upon landing	Landing 16. WILCO.5-SF	5-SF landing time 16. Report vacating runway.
Vacating the runway	Tower, 5-SF vacating the runway. Over. Apron Control on 125.7. 5-SF	5-SF Roger. Contact Apron Control 125.7. Over.

STANDARD PHRASEOLOGY FOR SSR COMMUNICATIONS

PHRASE	MEANING
"Squawk"	Operate transponder or check its operating condition
"Squawk Alpha Code 1027"	Select Mode A and Code 1027
"Squawk Three Code"	Select Mode 3 and Code (IFF/SIF equipped aircraft)
"Squawk Ident"	Use IDENT or I/P feature, retaining present code
"Squawk Alpha 7700"	Select "Emergency"
"Squawk Standby"	Switch to Standby retaining present code
"Squawk Altimeter"	Switch on altimeter reporting facility
"Stop Squawk"	Switch off transponder
"Stop Squawk Three"	Switch off Mode 3 (Military aircraft only)
"Stop Altimeter Squawk"	Switch off altitude reporting
"Recycle (Mode) (Code)"	To request the pilot to reselect the assigned mode and code

SOME USEFUL Q-CODES

QBD	=	Fuel endurance
QBI	=	Compulsory IFR conditions
QDL	=	Do you intend to ask me for a series of bearings?
QDM	=	Magnetic heading to steer to reach station in zero winds
QDR	=	Magnetic bearing fro station
QFE	=	Atmospheric pressure at aerodrome elevation (or runway threshold) or barometric pressure reduced to aerodrome elevation
QFU	=	Magnetic orientation of runway
QGE	=	What is my distance to your station? Or, your distance to my station is
QNH	=	Altimeter sub-scale setting to obtain elevation when on ground
QSY	=	Change of frequency
QTE	=	True bearing from station
QBO	=	Visual meteorological conditions (VMC) exist
QUAD	=	Quadrant
QUJ	=	Will you indicate the true track to reach you? Or, the true track to reach me is

OPERATION IN THE NAIROBI CONTROL ZONE

The Nairobi Control Zone is divided into three areas: (See attached Diagram)

- (a) AREA I: Ground to 9000 AMSL
 - Control Authority: (i) Wilson Tower when military zone is not active
 (ii) Eastleigh Tower when military zone is active; or
 (iii) Nairobi Approach when Wilson is closed
- (b) AREA II: Ground to 6500 AMSL
 - Control Authority: (i) Wilson Tower; and
 (ii) Nairobi Approach when Wilson AD is closed.

(NB: Terrain clearance not provided in AREA II)
- (c) AREA III: Ground to 9000 AMSL and also area above AREA II to 9000 AMSL
 - Control Authority: Nairobi Approach.

11.7 VFR Routes

- 11.7.1 **NE Access Lane:**
 - (i) Between Eastleigh and JKI Aerodromes.
 - (ii) Track 069° M from Wilson to ABM THR 06 HJK then 056° M to Ndula Marker.
 - (iii) Outbound aircraft to fly at 6200 ft & inbound at 6500 ft. AMSL
 - (iv) Pilot must request permission form JKIA Approach frequency 119.7 MHz.
- 11.7.2 **SE Access Lane:**
 - (i) Track 324° M from twin Bridges to Visual markers (4.6NM from Wilson bearing 196°M)
 - (ii) Outbound fly at 6200 ft and inbounds at 6,500 ft AMSL
 - (iii) Clearance must be obtained form Nairobi Approach on 119.7 MHz.

SELF STUDY QUESTIONS

1. What is the minimum age for the grant of a R/T licence? *[17 Years]*
2. Who is responsible for the issue of a flight radiotelephony operator's licence to persons who qualify? *[DG KCAA]*
3. Who is the authority for licensing of aircraft radio stations?
[Commissioner General, Communications Commission of Kenya]
4. What are the application procedures for the R/T licence?
 - [(i) Provide proof of having passed R/T Licensing Examination,*
 - (i) Complete application forms obtainable from the licensing section of the KCAA No medical examination is required.*
 - (ii) Photos:- Complete application forms forwarded to licensing section with two un-mounted photographs of the head (1¼" x ¾")]*
5. Which class of medical examination is required for the issuance of a R/T licence an applicants? *[None]*
6. In which document would one refer to for fees charged in respect of R/T Operator's licence? *[AIC Kenya]*
7. What are the privileges of a R/T licence holder? *[To use the radiotelephony on board an aircraft]*
8. Who are exempted from the requirement to sit for R/T licensing examination?
[NB: Not in the KCARS, but in CAP 90:
 - (i) Applicants who provide evidence of possession or equivalent qualification.*
 - (ii) Holders of a Kenya ATC Licenses.]*
9. What are the application procedures for a radiotelephony licence?
 - [(i) Must have passed RTF examination.*
 - (ii) Fill application form obtained at KCAA, Licensing.*
 - (iii) Submit passport size photographs]*
10. What are the conditions to be met in order to convert a foreign pilot's licence to a Kenyan one with respect to R/T licence?
[If the licence does not include RTF Operator's Licence, the applicant must sit for and pass a RTF examination before such licence].

11. What is the maximum validity period for a R/T licence? *[24 Months]*
12. What are the R/T licence renewal requirements? *[Holder must have exercised the privileges of the licence in the six months preceding the date of application]*
13. Explain the difference between an altitude and a flight level.
[An altitude is a vertical distance of a level, a point or an object considered as a point, measured from mean sea level, while a flight level is surface of constant atmospheric pressure which is related to a specific datum, 1013.25 hPa, and is separated from other such surfaces by specific pressure intervals].
14. Flying in the eastern Africa region, when would a departing aircraft change from the QNH setting on the aircraft altimeter's sub-scale to 1013.25?
[On reaching the Transition Altitude]
15. Define:
 - (i) Aeronautical station. *[A land station in the aeronautical mobile service. (In certain instances, it may be located on board a ship or on a platform at sea).]*
 - (ii) Aeronautical mobile service *[A mobile service between aeronautical stations and aircraft stations, or between aircraft stations]*
 - (iii) Aeronautical Fixed Service *[A telecommunication service between specified fixed points provided primarily for the safety of air navigation and or the regular efficient and economical operation of air services].*
16. Which altimeter setting is used in the following circumstances?
 - (i) Take-off and climb: *[QNH setting until the transition altitude is reached]*
 - (ii) En-route: *[1013.25 hPa and aircraft flown at flight levels at all times during IFR].*
 - (iii) Approach and Landing: *[On penetrating the transition level the QNH setting will be used]*
17. Which units of measurements are used in the provision of air traffic services in Kenya?

[The SI Units in conjunction with some Non-SI Units such as feet, nautical miles and degrees Centigrade]

18. Which units of dimensions are used in the determination of the following quantities?
- (i) Horizontal speed including wind speed *[Knots/Km per hour]*
 - (ii) Altimeter setting *[HectoPascal]*
 - (iii) Weight *[Metric Tonnes or Kilograms].*
19. What is an aerodrome traffic circuit?
[The specified path to be flown by aircraft operating in the vicinity of an aerodrome]
20. Draw a sketch of a standard aerodrome traffic circuit.
[See diagram at 2.10]
21. In which direction should aircraft turn after take-off when flying in a standard aerodrome traffic circuit? *[Left]*
22. What do the words "Land after" used by an ATC Unit mean? *[Land at your discretion keeping the preceding aircraft in sight]*
23. What are conditions to be observed for the words "Land after" to be used?
*[(i) The length of the runway permits;
(ii) It is during daylight hours; and
(iii) The following aircraft will be able to see the preceding one clearly and continuously until it has turned off the runway.]*
24. When may departing aircraft be cleared for take-off or an arriving aircraft cleared to land in respect of a preceding departing aircraft?
*[(i) is airborne and has retracted its undercarriage; or
(ii) (in the case of a fixed undercarriage aircraft), is airborne and has cleared the upwind end of the runway in use.]*
25. In what form is the aeronautical information provided in Kenya?
[Aeronautical information is provided in the form of an Integrated Aeronautical Information Package]
26. What are the contents of the aeronautical information package?
*[(i) Aeronautical Information Publication (A.I.P)
(ii) Amendment Service to the AIP (A.I.P. AMDT)]*

- (iii) *Supplement to the A.I.P (AIP SUP)*
- (iv) *NOTAM and Pre-Flight Information Bulletin (P.I.B)*
- (v) *Aeronautical Information Circulars (A.I.C)*
- (vi) *Checklist and Summaries]*

27. Which is the basic aviation document intended to satisfy international requirements for exchange of aeronautical information? *[The AIP]*
28. What type of aeronautical information is contained in the AIP?
[Permanent aeronautical information and long duration temporary changes essential for air navigation]
29. What are the two types of AIP amendment services?
[Regular AIP Amendment (AIP AMDT) and AIRAC AIP Amendment (AIRAC AIP AMDT)]
30. Which aeronautical publication contains information of short duration consisting of extensive graphics additional to the Aeronautical Information Publication?
[Supplements to the AIP; (AIP SUP)]
31. How the aeronautical information the timely knowledge of which is essential to personnel concerned with flight operations disseminated?
[Via the AFTN in form of NOTAM]
32. What is the difference in the distribution of NOTA Series A and NOTAM Series B?
*[(i) Series A: This is given National and International distribution.
(ii) Series B: This is given International distribution to adjacent FIR only.]*
33. How frequent are the validity checklists of the following publications issued?
- (a) NOTAMS *[Monthly]*
 - (b) AIC *[Once every year]*
34. Describe the AIRAC System.
*[The System consists of effective dates on 28-day cycles and the information is issued so that it reaches the user:
(i) 28 days before the effective date, or
(ii) 56 days before the effective date for major changes.]*
35. In what form is electromagnetic energy propagated?
[Wave form]

36. Define the following terms in regard to radio waves:
- (a) Wavelength *[Distance traveled by the wave energy in the direction of propagation during one cycle]*
 - (b) Frequency *[Number of cycles which pass a fixed point in one second]*
37. What is the speed of radio waves?
- [(i) 162 000 Nautical Miles per Second, or*
 - (ii) 300 000 000 Metres per Second; or*
 - (iii) 186 000 Statute Miles per Second; or*
 - (iv) 300 000 Kilometres per Second]*
38. State the relationship between a radio wave frequency and its associated wavelength? *[Wavelength is inversely proportional to the frequency]*
39. Given a frequency of 345 KHz. what will be its wavelength? *$[\lambda = c/f$
 $= 300 \times 10^6 / 345 \times 10^3 = 870 \text{ metres}]$*
40. What is the frequency of a radio wave whose wavelength is 30 cm?
 $[f = c/\lambda = 300 \times 10^6 / 0.3 = 1,000,000,000 \text{ Hz.} = 1 \text{ GHz.}]$
41. In which frequency bands are the following frequencies?
- (i) 118.0 MHz. *[VHF]*
 - (ii) 350 KHz. *[MF]*
 - (iii) 288 GHz. *[EHF]*
 - (iv) 249 KHz. *[LF]*
 - (v) 18899 KHz. *[HF]*
42. Name the classifications of radio wave propagation.
- [(i) Ground waves*
 - (ii) Sky waves*
 - (iii) Direct waves]*
43. What do you understand by the term "wave diffraction"?
[The bending of transmitted radio waves enabling them to travel close to the earth's surface.]
44. How does diffraction of radio waves vary with changes in wavelength?
[Diffraction decreases as wavelength is decreased]
45. What is the minimum radio equipment required by law to be carried by all aircraft flying under VFR? *[A radio communication equipment capable of conducting two-way communication with those aeronautical stations and on the frequencies prescribed by the Authority.]*

46. Define:
- (a) Aeronautical Station
[A land station in the aeronautical mobile service. In certain instances, it may be located on board a ship or on a platform at sea.]
 - (b) Aeronautical Mobile Service
[A mobile service between aeronautical stations and aircraft stations, or between aircraft stations.]
 - (c) Transition layer.
[The airspace between the transition altitude and the transition level]
 - (d) Air-to-ground communication.
[One way communication from aircraft to stations or locations on the surface of the earth]
 - (e) Air-ground communication.
[Two way communication between aircraft and stations or locations on the surface of the earth]
 - (f) Broadcast.
[A transmission of information relating to air navigation that is not addressed to a specific station]
47. What is a pilot required to do if it becomes necessary for an aircraft station to send signals for testing or adjustment of radio apparatus?
[The concord of the aeronautical station must be obtained and such transmission shall be kept to a minimum.]
48. Whose responsibility is it to establish communication in the aero-mobile service?
[The station having traffic to transmit.]
49. When making calls to an aeronautical station, what is recommended time interval between any two successive calls?
[10 seconds]
50. When an aeronautical station is called simultaneously by several aircraft stations who shall decide the order in which the aircraft shall communicate?
[The aeronautical station.]
51. What are the categories of messages handled by the aeronautical mobile service? Give them in the order of priority.
- [(a) Distress calls, distress messages & distress traffic*
 - (b) Urgency messages, including messages proceeded by the medical transports signal*
 - (c) Communication s relating to direction finding*
 - (d) Flight safety messages*
 - (e) Meteorological messages*
 - (f) Flight regularity messages]*

52. In which category of messages is the meteorological advice of immediate concern to an aircraft in flight or about to depart?
[Flight safety category]
53. List the types of messages within the Flight Safety Category with regard to aero-mobile service communications.
*[(i) Movement and control messages.
(ii) Messages originated by aircraft operating agency or by aircraft of immediate concern to an aircraft in flight.
(iii) Meteorological advice of immediate concern to an aircraft in flight or about to depart.
(iv) Other messages concerning aircraft in flight or about to depart.]*
54. In which order should messages having same priority be transmitted?
[In the order in which they are received for transmission.]
55. What is the objective of standard radiotelephony phraseology?
[To give both speed and intelligibility and because they are familiar to all users, will be more readily understood in difficult operating conditions than non-standard phraseologies]
56. How are the following transmitted on R/T?
- | | | |
|-------|--------------|--|
| (i) | 2,002 | <i>[TOO ZERO ZERO TOO]</i> |
| (ii) | 1020 UTC | <i>[WUN ZERO TOO ZERO]</i> |
| (iii) | 45,000 | <i>[FOWER FIFE TOUSAND]</i> |
| (iv) | 119.245 MHz. | <i>[WUN WUN NINER DAYSEEMAL TOO FOWER]</i> |
57. How are aeronautical stations in the aeronautical mobile service identified?
*[(i) The name of the locations, and
(ii) The unit or service available.]*
58. List the information a pilot is required to read back in R/T procedures.
*[(i) ATC route clearances
(ii) Clearances and instructions to enter, land on, take-off on, hold short of cross or backtrack any runway
(iii) Runway in use, altimeter settings, SSR Codes, level instructions, heading and speed instructions]*
59. What would be the correct read backs to the following phrases:
- (i) "5Y-MMB, Wilson Tower, cleared to land"
[Cleared to land. 5Y-MMB]

- (ii) "5Y-MNC, Wind 065/10 taxi holding point RWY 07. QNH 1024 hectopascals"
[Holding point RWY 07. QNH 1024. 5Y-MNC]
- (iii) "5Y-KSF, Contact Area Control on 118.5. over." *[5Y-KSF]*
- (iv) "5Y-ANC. Land after the Cessna 310 just landed on RWY 07"
[5YANC]
60. 73. State the procedure for sending signals for testing or adjustment of radio equipment.
 [(i) Identification of the station being called
 (ii) Aircraft identification
 (iii) The words "RADIO CHECK"
 (iv) The frequency being used.]
61. On the radio readability scale, what does strength three signify?
[Readable but with difficulty]
62. On the radio readability, what strength is given to a transmission that readable now and then?
[Strength 2]
63. What is the approximate northern limit of the Nairobi FIR? *[04°N]*
64. What are the Nairobi ACC VHF frequencies? *[120.5MHz.; 121.3MHz.; 118.5MHz.; 125.3MHz.]*
65. What is the VHF Cover Scheme Frequency for aircraft operating to west of Nairobi?
[118.5 MHz.]
66. What is the VHF emergency frequency?
[121.5 MHz.]
67. What are the Nairobi FIR HF primary frequencies? *[11300 KHz. (Day) & 5517 KHz. (Night)]*
68. What do you understand by the term "Regular Station"?
[A selected station among a large number serving a given route segment]
69. What is the main factor considered in selecting a regular station?
[Propagation characteristics of the frequencies]
70. State the three responsibilities of a regular station.
 [(i) Designating suitable primary and secondary frequencies for its communication with the aircraft

- (ii) *Receiving all position reports and handling other messages from and to aircraft essential to the safe conduct of flight.*
- (iii) *The action required in case of failure of communications]*

71. On which frequency should an aircraft operating at unmanned aerodrome broadcast its intentions? *[118.0 MHz.]*
72. What are the recommended limits within which an aircraft operating into unmanned aerodromes should make communication broadcasts?
[Not outside a radius of 10 nm from the aerodrome or at greater height than 3000 ft above ground level]
73. What is the significance of aircraft operating into unmanned aerodrome broadcasting within the limits in Q.71?
[To minimize interference with communications of aircraft operating into adjacent aerodromes]
74. How does transmitting blind differ from broadcasting?
[In blind transmission two-way communication cannot be established but it is believed that the called station is able to receive the transmission, whereas in a broadcast, the transmitted information is not addressed to a specific station or stations.]
75. What is an Air Report?
[A report from an aircraft in flight prepared in conformity with requirements for position, operations and/ or meteorological reporting.]
76. How are Air Reports made by R/T to be transmitted?
[They are transmitted in AIREP Forms]
77. When are aircraft flying on routes not defined by designated significant require to make air reports?
[As soon as practicable after transfer of control has taken place and at hourly intervals thereafter.]
78. What is the objective of "Operations Normal" reports? *[Applied with the object of facilitating SAR]*
79. When should a pilot on a VFR flight transmit position reports?
[VFR lights shall make RT contact with appropriate ATS Units on the en-route frequency giving a position report as follows:
 - (i) *As soon as possible after departure.*
 - (ii) *When changing frequency*
 - (iii) *When destination is in sight*

(iv) *On flights of sufficient duration, an "Operations Normal" call or position report shall be made at intervals of not more than one hour.]*

80. What are the sections of an air report?

- [(a) SECTION I Position Report*
- (b) SECTION II Operating Information*
- (c) SECTION III Meteorological Information]*

81. Which items of an Air Report are to be transmitted to an ATS Unit on first contact where a Flight Plan had not been filed?

[All of Sections I and II]

82. Name the designated meteorological reporting points in the Nairobi FIR.

- [(i) The boundaries of the FIR.*
- (ii) At hourly intervals when flying over the Indian Ocean at a distance of 30 minutes' flying time or greater from the shore irrespective of the flight level.*
- (iii) At any time when the observed weather differs significantly from the forecast given.*
- (iv) Pilots are requested to pass to an appropriate ATS Unit observations on adverse weather on climb out or approach to land.]*

83. When are Special Air Reports required?

[Special Air Reports shall be made by all aircraft on all routes whenever:

- (i) Severe icing or turbulence is encountered; or*
- (ii) Other meteorological conditions are encountered which in the opinion of the pilot in command are likely to affect safety of other aircraft.].*

84. When shall Additional Reports be made?

[Additional air reports shall be made if a specific request is made before or during flight by a meteorological service for the flight.]

85. What is the significance of a distress message?

[To alert other stations of existence of an emergency situation so that the traffic is afforded due priority]

86. What is the difference between a distress message and an urgency message?

[A distress message indicates a condition of being threatened by serious and/or imminent danger and requiring immediate assistance, while an urgency message indicates a condition concerning the

safety of an aircraft or other vehicle, or some person on board or within sight, but which does not require immediate assistance].

87. What are the commencing words for an urgency message transmission?
[PAN PAN PAN]
88. What does the R/T signal "PAN PAN PAN" indicate?
[Commencement of an urgency message]
89. On which frequency should distress or urgency messages first be made?
[First call should be to the ATS Unit with which it has already established communication or to one which is known to be within range].
90. List the frequencies which are available for use by an aircraft in emergency.
*[(i) 121.5 MHz (Emergency Locator Beacon –Aircraft: ELBA); or
(ii) 243 MHz (ELBA for Military); or
(iii) 500 KHz (International distress frequency); or
(iv) 2182 KHz (Maritime Mobile Distress Frequency)]*
91. What is "ELBA" and what is its purpose?
*[ELBA = Emergency Locator Beacon –Aircraft: ELBA
Purpose: Assist in location of an aircraft crash site]*
92. On what frequency does "ELBA" operate? *[121.5 MHz.]*
93. What is the correct phrase for the imposition of silence in a distress situation?
["STOP TRANSMITTING"]
94. What is the correct phrase for terminating silence conditions imposed during distress communication?
["DISTRESS TRAFFIC ENDED"]
95. How are aircraft emergency phases classified?
*[(i) Uncertainty Phase (INCERFA)
(ii) Alert Phase (ALERFA)
(iii) Distress Phase (DETRESFA)]*
96. Define each of the aircraft Emergency Phase.
*[(i) **Uncertainty Phase (INCERFA)**: – a situation where in uncertainty exists as to the safety of an aircraft and its occupants.
(ii) **Alert Phase (ALERFA)**: - A situation wherein apprehension exists as to the safety of an aircraft and its occupants.*

(iii) **Distress Phase (DETRESFA):** - A situation wherein there is reasonable certainty that an aircraft and its occupants are threatened by grave danger and require immediate assistance.]

97. State the circumstances that would prompt uncertainty phase.
- [(i) No communication has been received from an aircraft within a period of thirty minutes after time a communication should have been received, or the time unsuccessful attempt to establish communication with such aircraft was first made, whichever is the earlier; or
 - (ii) An aircraft fails to arrive within thirty minutes of estimated time of arrival last notified to or estimated by air traffic services units, whichever is the later, except when no doubt exists as to the safety of aircraft and its occupants.]
98. Which phase of emergency will the ATS enter directly if an aircraft cleared to land does not do so within five minutes?
- [If an aircraft has been cleared to land and fails to land within five minutes of the estimated time of landing and communication has not been re-established with the aircraft.]*
99. Which emergency phase will an ATS unit declare in the event of an aircraft being known or believed to be the subject of unlawful interference.
- [Alert phase; i.e. "ALERFA"]*
100. Where are the following SAR Units located within the Nairobi FIR?
- (i) The Rescue Coordination Centre (RCC) *[Adjacent to the ACC at JKIA]*
 - (ii) Type I Rescue Sub-Centres *[At ATC Units countrywide]*
 - (iii) Type II Rescue Sub-Centres *[At Provincial Police HQS.]*
101. Which ground visual signal would survivors of aircraft crash use to indicate the following?
- (i) Medical assistance required [**X**]
 - (ii) Assistance in general required [**V**]
102. What do the letters "IBRD" represent?
- [International 406 MHz. Beacon Registration Database]*
103. Give a brief description of the "IBRD" System operation.

[Supports element of the COSPAS- SARSAT alert system that detects and distributes signals transmitted by 406MHz. Emergency Locator Transmitters]

104. How do Direction Finding station within a region work?
[The work either singly or in groups under the direction of a main direction-finding station.]
105. What do the following codes used in DF. Mean?
- (i) QTE *[True bearing of the aircraft from the station]*
 - (ii) QUJ *[True heading to be steered by aircraft, with no wind, to head for the direction finding station.]*
 - (iii) QDR *[Magnetic bearing of aircraft, form the station.]*
 - (iv) QDM *[Magnetic heading to be steered by aircraft, with no wind, to make for the station.]*
106. Which DF station will give an aircraft position in relation to a point of reference or in latitude and longitude using the signal QTE?
[A station controlling a network where the DF stations are working in a group]
107. How are DF stations classified?
[They are classified in accordance with their bearing accuracy; thus:
- | | | |
|---------|------------------|----------------------------|
| Class A | $\pm 2^{\circ}$ | <i>within 9.3 km (5nm)</i> |
| Class B | $\pm 5^{\circ}$ | <i>within 37 km (20nm)</i> |
| Class C | $\pm 10^{\circ}$ | <i>within 92 km (50nm)</i> |
| Class D | < Class C | <i>Less than Class]</i> |
108. What does QDM 150 B mean? *[Assuming zero wind condition, Steer 150°(M) $\pm 5^{\circ}$ to reach station]*
109. Who is the Authority responsible for the provision of aviation meteorological services?
[The Director of Meteorological Services in the MOT]
110. Which ICAO Document contains Standards and Recommended Practices for civil aviation meteorological services?
[ICAO Annex 3]
111. At which stations in Kenya are meteorological observations and reports made?
[At all stations where ATC is provided]

112. Where should a pilot obtain route weather, area and landing forecast for an unmanned station?
[From any aerodrome units on notified telephone numbers]
113. How much time before departure should a pilot request for route weather forecasts?
[At least 6 hours]
114. List the elements of a routine weather report passed to aircraft in the correct order.
- [(i) Station and time of report;*
 - (ii) Surface wind, direction and speed*
 - (iii) Horizontal visibility (or RVR reading when applicable)*
 - (iv) (Present weather if significant)*
 - (v) Cloud amount and base*
 - (vi) Temperature and Dew point*
 - (vii) QNH (or QFE on request)*
 - (viii) Supplementary information in plain language.]*
115. In meteorological reporting, what does the term 'CAVOK' mean?
[conditions of clouds and visibility okay; i.e. better than prescribed values]
116. How is the receipt of a routine meteorological report acknowledged?
[Acknowledging receipt of the weather elements, reading back the QNH setting and terminating by its call sign]
117. What are the correct radiotelephony for the following?
- (i) "Yes" *[AFFIRM]*
 - (ii) "All aircraft stations on this frequency" *[ALL STATIONS]*
 - (iii) "Examine a system or procedure" *[CHECK]*
 - (iv) "Pass the following information" *[REPORT]*
 - (v) "I hereby indicate the separation between messages transmitted to the different aircraft in a very busy environment." *[BREAK BREAK]*
118. What are the correct phraseologies for the following?
- (i) "Check and confirm with originator." *[VERIFY]*
 - (ii) "Reduce your speech rate." *[SPEAK SLOWER]*
 - (iii) "Authorized to proceed under conditions specified." *[CLEARED]*
119. What do the following phraseologies mean when used in R/T communication?

- (i) CHECK *[Examine a system or procedure. (No answer is expected).]*
 - (ii) SAY AGAIN *[Repeat all, or the following part of your last transmission.]*
 - (iii) DISREGARD *[Consider that transmission as not sent].*
 - (iv) VACATE LEFT *[Turn left to leave the runway]*
120. How the separation between messages transmitted to two different aircraft in a very busy environment indicated? *[BREAK BREAK]*
121. How are the following units identified on R/T?
- (i) Area Control Centre *[CONTROL]*
 - (ii) Approach Control *[APPROACH]*
 - (iii) Flight Information Service *[INFORMATION]*
 - (iv) Aerodrome Control *[TOWER]*
122. What do the following Q-codes refer to?
- (i) QDR *[Magnetic bearing from station]*
 - (ii) QGE *[What is my distance to your station? Or, your distance to my station is]*
 - (iii) QDL *[Do you intend to ask me for a series of bearings?]*
 - (iv) QFU *[Magnetic orientation of runway]*
 - (v) QBI *[Compulsory IFR conditions]*
123. What is the Q-code for the following?
- (i) Fuel endurance *[QBD]*
 - (ii) Change of frequency *[QSY]*
 - (iii) Magnetic orientation of runway *[QFU]*
 - (iv) Barometric pressure reduced to aerodrome elevation *[QFE]*
 - (v) Visual meteorological conditions (VMC) exist *[QBO]*